

Refinements to Rceattle for eastern Bering Sea pollock

Rceattle working paper for the 2024 assessment configuration

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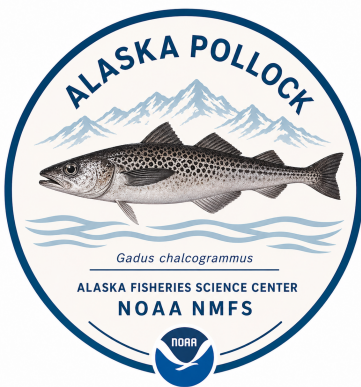
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Abstract

The custom AD Model Builder (ADMB) stock assessment model developed for eastern Bering Sea walleye pollock has a long history of application. For a variety of reasons, including general recommendations from the 2025 Center for Independent Experts (CIE) reviews, the custom ADMB code has been successfully ported to R Template Model Builder (RTMB). That conversion provides access to advanced RTMB features and makes the underlying code easier to understand and modify. Similarly, adopting more generic platforms, such as Rceattle, can provide access to more modern software developments. In this study, we tailored Rceattle code and configurations so that Rceattle can closely mimic the custom ADMB/RTMB assessment code. The Rceattle configuration uses the supported multinomial composition likelihood with a robustifying constant and a non-parametric selectivity form. We also evaluated a two-dimensional, first-order autoregressive (AR1) selectivity sensitivity; it did not converge and remains a research topic. An exploratory dynamic structural equation modeling (DSEM) analysis shows how recruitment covariates can be compared by cohort, separately from the final assessment configuration. A controlled migration check under Rceattle 5.20.0, pinned to the tested source revision, exactly reproduced the accepted objective, gradient, and principal population trajectories.

AI-assisted development

Artificial intelligence (AI) tools were used to aid document development and model implementation. AI-assisted material was reviewed and revised by the authors, who retain responsibility for the scientific content, code, analyses, interpretations, and conclusions.



Rceattle WORKING PAPER

1 Executive Summary

Building from the ADMB-to-RTMB conversion, this analysis evaluates whether Rceattle can be configured to reproduce the main features of the 2024 eastern Bering Sea pollock “PM” assessment while providing a model

that is easier to extend. This update focuses on the Rceattle configuration, comparison with the modified ADMB model, model diagnostics, and projection inputs.

When the Rceattle parameters and fishery selectivity are set to the same values used in the ADMB model, Rceattle reproduces numbers-at-age, spawning biomass, and predicted catch to numerical precision. Mean absolute differences are 0.000117% for spawning biomass and 0.000143% for predicted catch. This result establishes agreement in the population dynamics when both implementations use the same parameter values.

When Rceattle was fit to the data, the convergence criteria were fully met, including a positive-definite Hessian. The least-identified direction remained in annual ATS/AVO selectivity deviations, but the added direct coefficient-increment penalty reduced the condition number below the warning threshold. Relative to the ADMB fit, mean absolute differences were 1.5% for spawning biomass, 0.7% for recruitment, and 10.3% for total biomass. These small differences are concentrated in the earliest years; the remaining uncertainty is primarily estimability of the annual selectivity deviations.

The 2D age-by-year AR1 selectivity sensitivity was included as an initial diagnostic attempt. Results appeared broadly similar to the non-parametric constrained selectivity fits. However, because some parameters reached bounds, the model failed to converge. Closer examination of this selectivity form will continue as a research topic.

One-step-ahead residuals and nine terminal-year-consistent retrospective peels have been generated from the Rceattle configuration. Aggregate composition residual dispersion is lower than the standard-normal reference. Mohn's rho is 0.261 for spawning biomass, 0.119 for total biomass, 0.147 for recruitment, and -0.077 for fishing mortality. Forty-eight of 50 jitters returned to the best solution; two found a substantially poorer mode. These results focus subsequent work on selectivity identification, retrospective bias, and observation-model sensitivity.

2 Purpose and Scope

The September 2026 review focuses on whether the Rceattle configuration has sufficient documentation for Plan Team evaluation. This working paper serves that focused purpose rather than the broader purpose of a full Stock Assessment and Fishery Evaluation (SAFE) chapter.

This working paper therefore records:

- what was run from those development files;
- how the Rceattle outputs compare with the modified ADMB reference;
- which diagnostics are available now;
- which sensitivities remain research-only; and
- what work would help structure the next evaluation pass.

The report uses the [Alaska Stock Assessment Report \(asar\) package](#) for glossary conventions and accessibility guidance ([Schiano et al. 2026](#)). Those practices helped prepare the Quarto source for Section 508 review, including semantic headings and tables, descriptive figure captions, and alternative text. They support accessibility but do not replace final manual and agency-approved compliance checks.

Standard SAFE report sections such as background material, fishery history, ecosystem narrative, reference-point tables, and harvest advice are outside the scope of this document. Projection calculations are included only to test the export and use of the Rceattle results.

3 Rceattle Configuration

3.1 Model Inputs

The Rceattle configuration uses the prepared workbook `Data/EBS_24_pollock_m23_rceattle_full_1964-2024.xlsx`. The data builder retains every age from 1 through 15. Fishery compositions use nominal sample sizes; bottom trawl survey (BTS) and acoustic-trawl survey (ATS) sample sizes use the integer values aligned

with the modified ADMB model. Age-1 survey selectivity remains separately parameterized from the age-2–15 curve, and each likelihood contribution evaluates the complete 15-age composition. Compositions use the multinomial likelihood with a robustifying constant. We added a constant of 0.001 to both observed and expected proportions before evaluating the likelihood. This prevents $\log(0)$ and limits the influence of observations in bins with expected proportions near zero.

The run uses Rceattle version 5.8.1 from upstream commit 7bb694788012, with the model configuration synchronized to Rceattle-models commit 3dc99ddfbf9e.

The next-assessment update script review is complete, and execution awaits complete new-year observations. The script duplicates terminal records and places placeholders for new catch, composition, and survey observations. The script serves as a template until complete new-year observations are available; after population, it creates a complete assessment workbook.

3.2 Selectivity

The configuration uses `NonParametricPM` fishery selectivity with annual deviations. When fit to the data, the model reached a small maximum gradient. The Hessian condition number was approximately 489,000 and the convergence criteria were fully met. Its least-identified direction remained concentrated in `sel_coff_dev`, specifically the shared ATS/AVO block rather than the shared fishery/CPUE block.

The time-varying selectivity penalties use explicit standard deviations on log-selectivity random-walk increments. The fishery and its shared CPUE selectivity block use an SD of 0.500, ATS and AVO use 0.138, and BTS uses 1.000. ATS and AVO also use a direct coefficient-increment penalty (`Sel_curve_pen3 = 0.1`) to constrain the across-age common-mode direction that is removed by annual selectivity normalization. The fishery specification also retains the ADMB-matched curve-shape penalty weights (12.5, 1/60, and 1) and an average-selectivity centering weight of 10. These values are penalty coefficients rather than SDs.

The fitting sequence, composition-likelihood sensitivity, and 2D age-by-year AR1 development work are documented in the appendix.

The ADMB and Rceattle fishery selectivity curves have similar age patterns, while their annual variation differs most in the earlier years (Figure 1).

For the BTS, both implementations use a logistic selectivity curve for ages 2–15 and override age 1 with a separately estimated parameter. The age-1 term therefore does not lie on the fitted logistic curve. Its annual deviation is multiplicative on the age-1 log-selectivity parameter, and the CV on the year-to-year deviations is specified at 25%. Rceattle encodes this ADMB specification as `LogisticPM`; the plotted age-1 values are the realized annual selectivities after applying the separate parameter and its annual deviations.

The BTS age compositions contribute directly to estimation of this selectivity through the multinomial likelihood with a robustifying constant. For each of the 42 BTS composition years, predicted numbers at age are formed from population abundance at the survey timing, BTS selectivity at age, and the ageing-error matrix, and are then normalized across ages 1–15. Catchability cancels during this normalization, so the compositions primarily inform the age pattern of selectivity and the population age structure rather than the absolute survey scale. In particular, the observed age-1 proportion informs the separate age-1 selectivity parameter and its annual deviations; ages 2–15 inform the logistic slope, inflection, their annual deviations, and abundance at age.

For year y , the fitted contribution is

$$-w_y N_y \sum_{a=1}^{15} (o_{y,a} + c) \log \left(\frac{p_{y,a} + c}{o_{y,a} + c} \right),$$

where $o_{y,a}$ and $p_{y,a}$ are observed and predicted proportions, $c = 0.001$, and the fleet composition weight w_y is 1. The nominal BTS sample sizes are truncated to the integer values read by ADMB; they range from 48 to 329 and sum to 5,403 across years. No young- or old-tail accumulation is used, so age 1 remains in the likelihood as its own bin. Larger- N_y years exert more influence, while c stabilizes empty or near-empty bins.

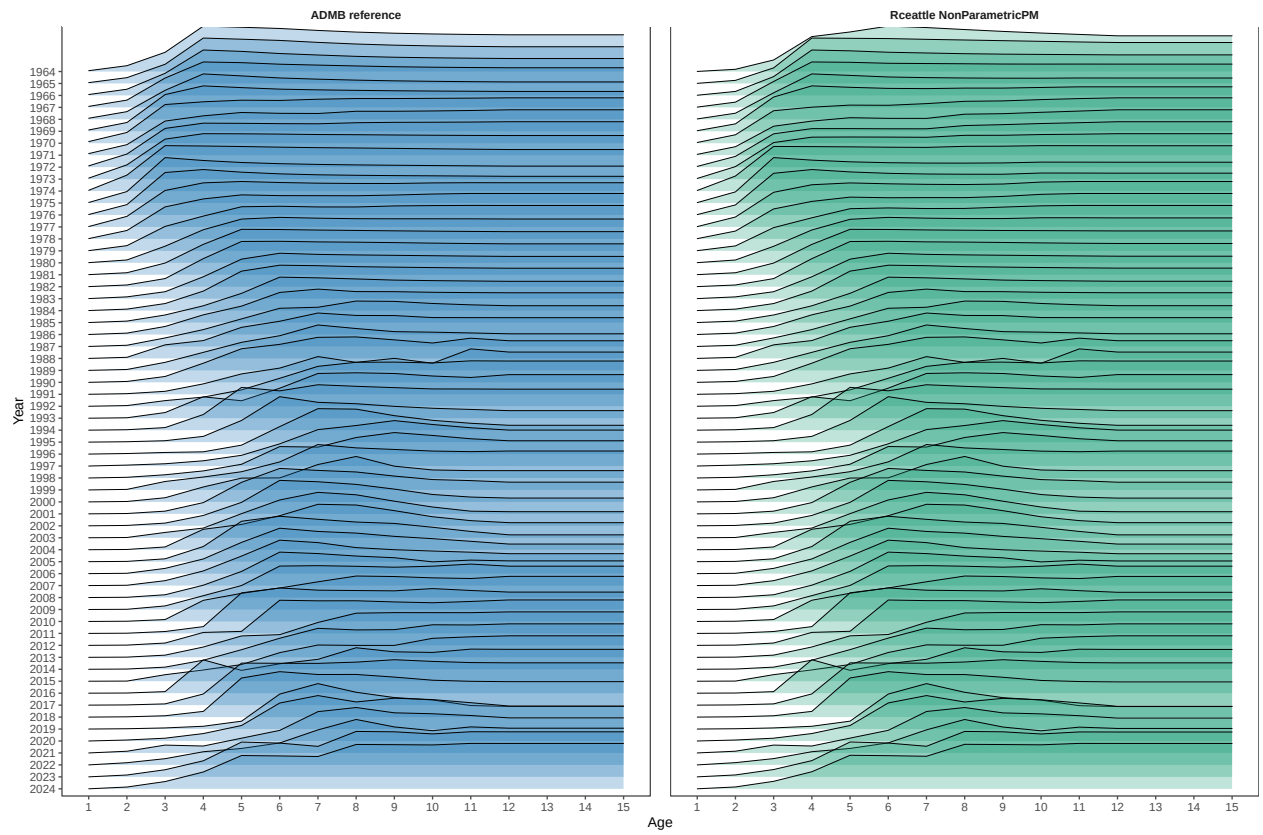


Figure 1: Relative fishery selectivity by year for the ADMB model and Rceattle configuration. Each annual curve is normalized to a maximum of one.

The composition contributions are optimized jointly with the BTS biomass index likelihood and the selectivity random-walk penalties; the fitted curve therefore balances annual age-composition fit, survey biomass fit, and temporal smoothness. Both implementations preserve the distinct age-1 term outside the logistic ages 2–15 curve (Figure 2).

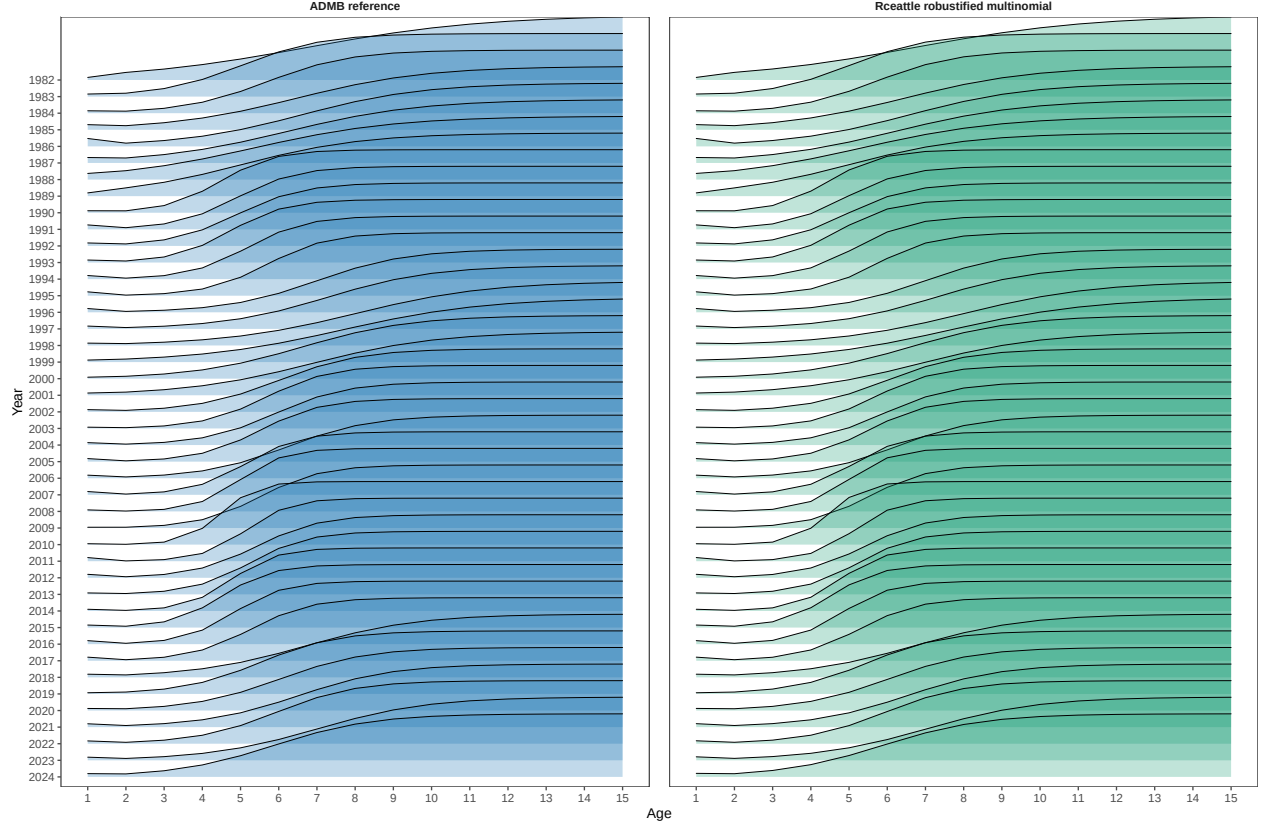


Figure 2: Relative bottom trawl survey selectivity by year for the ADMB model and Rceattle configuration. Ages 2–15 follow a time-varying logistic curve; age 1 is separately parameterized with annual random-walk deviations. Each annual curve is normalized to a maximum of one.

4 ADMB Comparison Results

When the Rceattle parameters and fishery selectivity are set to the ADMB values, the two programs give nearly identical population calculations. Mean absolute differences are 0.000117% for spawning biomass and 0.000143% for predicted catch. Annual Rceattle-to-ADMB ratios for both quantities remain extremely close to one (Figure 3).

The reported Rceattle likelihood contributions are 140.191 for indices, -123.761 for catch, 3572.874 for age compositions, and 98.379 for recruitment deviations. Their sum is 3704.588. The ADMB `tot_like` and `dat_like` values are 736.724 and 459.931, respectively, but cannot be compared directly because the two programs include different likelihood terms, penalties, and constants. The ADMB maximum absolute gradient is 3.47e-04; no Rceattle model-parameter gradient is available from this comparison because Rceattle does not estimate parameters in this run.

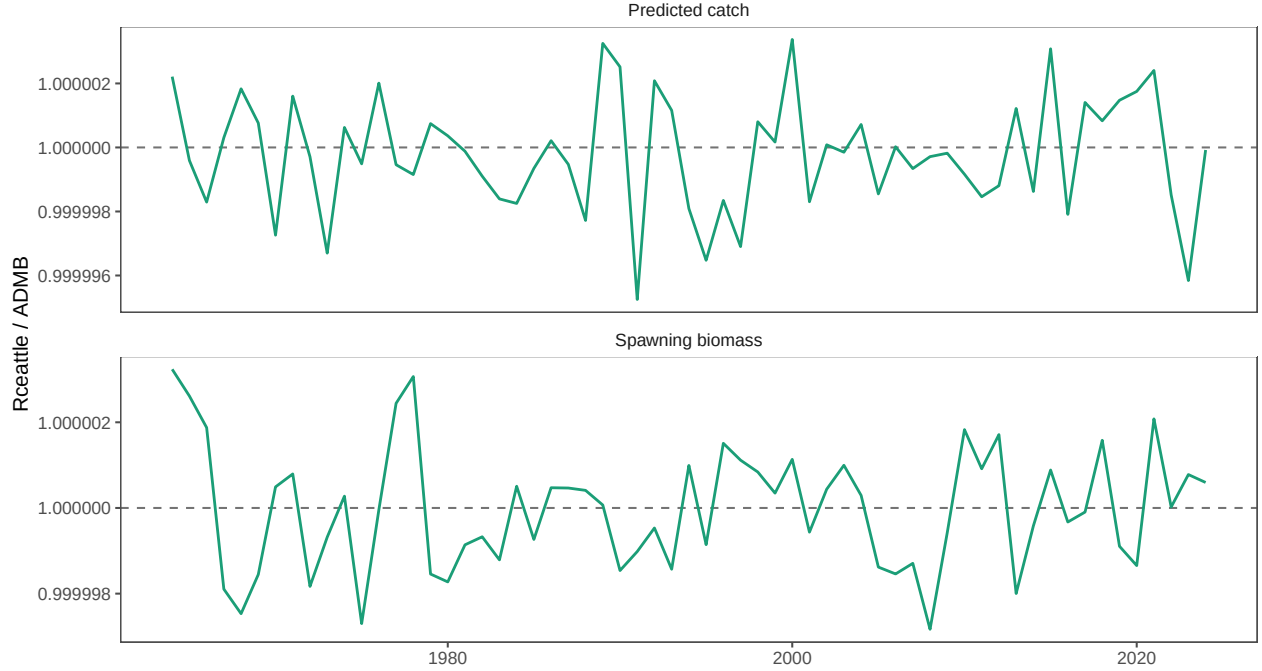


Figure 3: Rceattle-to-ADMB ratios for spawning biomass and predicted catch when Rceattle uses the ADMB parameter values.

Table 1: Agreement between fitted Rceattle trajectories and the modified ADMB reference. Correlation is the Pearson correlation across annual estimates. Mean absolute (%) and maximum absolute (%) are the average and largest absolute annual percentage differences, respectively. The 2024 difference (%) is the signed terminal-year difference of Rceattle relative to ADMB; negative values indicate that the Rceattle estimate is lower.

Quantity	Correlation	Mean absolute (%)	Maximum absolute (%)	2024 difference (%)
Rceattle configuration				
Recruitment	1.00	0.70	8.21	-4.97
Spawning biomass	1.00	1.45	9.96	2.88
Total biomass	0.98	10.32	95.63	2.61

When Rceattle estimates its parameters, the fitted trajectories differ most from ADMB during the early period and in the terminal recruitment estimate (Figure 4). These trajectories are distinct from the run in which Rceattle parameters are set to the ADMB values. The fitted trajectories remain highly correlated with the ADMB reference, with relatively small mean annual differences (Table 1). Development sensitivities are documented in the appendix.

The Rceattle fit has a small maximum gradient and a positive-definite Hessian. Adding the direct ATS/AVO coefficient-increment penalty reduced the condition number from approximately 1.14 million to 489,000 and removed the prior Hessian-conditioning warning. The residual weakest direction remains almost entirely in the shared time-varying ATS/AVO selectivity coefficients, not the fishery/CPUE coefficients, so this block remains an appropriate focus for sensitivity checks even though the formal warning is resolved.

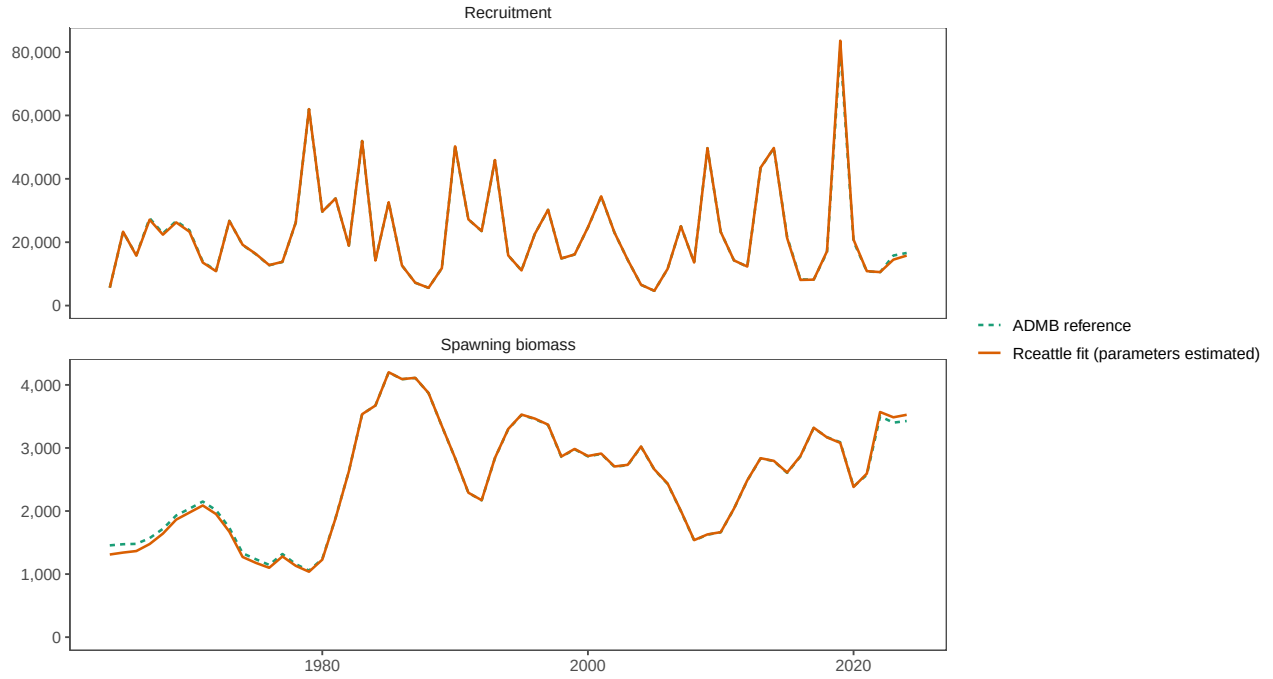


Figure 4: ADMB and fitted Rceattle trajectories for spawning biomass and recruitment. The Rceattle series is from the fit in which its parameters were estimated.

5 Diagnostics

The fitted Rceattle model fully met the convergence criteria: it reached a small maximum gradient, had a positive-definite Hessian, and remained below the Hessian-condition-number warning threshold. The supporting diagnostic products include OSA residuals, retrospectives, self-tests, and jitter tests (Table 2).

Table 2: Diagnostic status for the September 2026 review.

Diagnostic	Current status	Review use
ADMB-parameter comparison Rceattle fit	Completed and saved in the ADMB-to-Rceattle comparison outputs. Convergence criteria were fully met, with a positive-definite Hessian and a condition number of approximately 489,000.	Primary numerical agreement check. Primary Rceattle comparison fit to the data.
OSA residuals	Completed for 1,864 residuals and saved in RDS outputs.	Useful, but analytical and exactly reproduced index groups need interpretation.
Rceattle retrospectives	Nine peels completed and fully met the convergence criteria.	The revised peels preserve data availability and fix terminal fishery/CPUE selectivity increments at zero, holding the terminal selectivity-at-age curve at its preceding-year value.
Legacy ADMB retrospectives	Existing ADMB retro.rep files are present.	Provides historical context distinct from the Rceattle retrospective peels.
Rceattle 5.8.1 self-tests	Fifty initial-start and 50 estimated-start refits fully met the convergence criteria.	Supports the revised positive-index simulation workflow; it does not resolve observed-data multimodality.
Rceattle 5.20.0 migration	The canonical-schema workbook regenerated successfully and the final two-stage fit exactly reproduced the accepted 5.8.1 objective, gradient, and population trajectories.	Supports migration to the formal 5.20.0 release; the BTS selectivity and legacy Q_block messages remain soft-deprecation warnings.
Jitter tests	Forty-eight of 50 jitters returned to the best mode; two WARN fits reached an inferior mode.	Documents multimodality in the observed-data fit.

Table 3: Rceattle 5.8.1 self-test and paired-restart validation for the upstream-supported EBS configuration.

Metric	Value
Initial-start self-tests converged	5.000000e+01
Estimated-start self-tests converged	5.000000e+01
Paired fits with objective difference greater than 0.0001	0.00000e+00
Maximum absolute paired objective difference	3.09564e-08
Median across-simulation median SSB percent bias	-1.055%
Median terminal-year SSB percent bias	-0.907%
Fleets with at least one row above 2 percent rejection probability	2.00000e+00
Maximum row-level non-positive probability	6.66814e-02

Table 4: Fleet-level probability of a non-positive index draw under the fitted, untruncated observation distribution.

Fleet	Code	Distribution	Rows	Maximum row probability	Mean row probability	Rows above 2
ATS	4	Lognormal	19	0.00%	0.00%	
ATS_1	6	Lognormal	18	0.00%	0.00%	
AVO	2	Normal	18	3.88%	0.23%	
BTS	3	MVN	42	6.64%	0.17%	
BTS_1	5	Lognormal	42	0.00%	0.00%	
CPUE	7	Normal	12	0.00%	0.00%	

5.1 Rceattle 5.8.1 self-test validation

Rceattle 5.8.1 changed simulation of natural-scale normal and multivariate normal survey indices. The fitted likelihood remains an untruncated normal; the simulator redraws non-positive values because `data_check()` excludes them. The resulting self-test therefore uses a normal distribution truncated at zero whenever rejection occurs.

The ADMB-to-Rceattle comparison was rerun first with Rceattle 5.8.1. Its saved summary gives mean absolute differences of 0.000117% for spawning biomass and 0.000143% for predicted catch. This confirms that the package update preserved the agreement in the population calculations.

The self-test uses the same delivered EBS workbook and supported multinomial criterion with the same robustifying constant as the Rceattle configuration. The observed-data fit reproduced the stored objective of 716.948070 and maximum gradient of 0.000041; the convergence criteria were fully met, including a positive-definite Hessian. The Hessian condition number was approximately 4.9×10^5 ; its least-identified direction remained concentrated in the shared ATS/AVO `sel_coff_dev` block rather than fishery/CPUE.

All 50 phased simulations fully met the convergence criteria when refitted from the original initial values. Repeating the same simulated datasets from the fitted parameter values also produced 50 fits that fully met the criteria. Paired objectives agreed within 3.1×10^{-8} , so these simulations showed no start-dependent mode. Median spawning-biomass bias was -1.06% across years and -0.91% in the terminal year. These results validate the revised self-test workflow; they do not rule out the mode previously encountered when fitting the observed data (Table 3).

One AVO row had a 3.76% non-positive probability and one BTS multivariate-normal row had a 6.67% probability. Those rows exceed the package’s 2% warning threshold and should be identified when interpreting recovery. CPUE and the remaining index series had negligible or zero marginal rejection probability (Table 4).

5.2 Rceattle 5.20.0 migration check

The source and workbook workflow were also tested with Rceattle 5.20.0 at source revision bd9d0fcfe84a. The input-preparation script now writes the canonical 5.20.0 fleet-control names, explicitly specifies `MultinomialAFSC` and the index distributions, and passes the package’s strict data check. A two-stage refit from that regenerated workbook retained status OK, the objective 716.948070, and maximum gradient 4.14e-05. These values exactly match the accepted 5.8.1 fit, as summarized with the other diagnostics in Table 2.

The objective difference and all reported mean, maximum, and terminal percentage differences for spawning biomass, recruitment, and total biomass were 0e+00. The remaining messages concern the soft-deprecated BTS time-varying selectivity form and an ignored legacy `Q_block` field; neither changed the retained fit. The test therefore supports the pinned 5.20.0 source revision, which is the formal 5.20.0 release tag. Three simulation/refit smoke tests also converged, the five deterministic Rceattle-to-spmR input files had matching SHA-256 hashes, and all 98,000 rows of the seven-alternative projection output matched the established output.

5.3 OSA Residuals

OSA residuals run on the Rceattle configuration. The aggregate composition fits are generally close, while all three OSA quantile plots are narrower than the standard-normal reference (Figure 5). These diagnostics apply to the fitted `NonParametricPM` selectivity configuration. The failed 2D-AR1 sensitivity and ADMB-to-Rceattle calculation use separate results. The composition diagnostics use `afscOSA` version 0.0.1 and pass Rceattle’s internally calculated residuals to `afscOSA::run_osa()`. The residual builder uses complete ages 1–15, with nominal fishery sample sizes and the integer BTS and ATS sample sizes aligned with the modified ADMB model. This follows the current [afscOSA interpretation guidance](#) while retaining Rceattle’s residual calculation.

Following that guidance, the aggregate age-composition fits are the first check for systematic misfit, including patterns that could indicate an unsuitable selectivity form. The red vertical ranges contain 95% of data simulated from the fitted multinomial model conditional on the aggregate sample size and estimated parameter values held constant. They represent conditional simulation variability. The quantile–quantile plots then assess whether the OSA residuals resemble a standard normal distribution. The standard deviation of normalized residuals (SDNR) should be near one, and the displayed tail statistics provide a separate check on extreme residual behavior. Finally, the OSA and Pearson bubble plots are used to look for age, year, or cohort patterns that may suggest the source of misfit. Each statistic and nominal interval contributes to an integrated diagnostic interpretation rather than an automatic model-rejection rule.

The aggregate observed and fitted compositions are generally close, although the survey panels contain some age-specific departures. The OSA distributions are narrower than the standard-normal reference for all three composition sources: SDNR is 0.64 for the fishery, 0.80 for BTS, and 0.53 for ATS. Their tail diagnostics also show shorter-than-expected tails. Together these results indicate underdispersion relative to the fitted multinomial observation model. Model selection requires this information alongside the full suite of diagnostics. The bubble plots should instead be used to identify persistent age, year, or cohort patterns and to guide focused checks of selectivity, composition weighting, and observation-model assumptions. One Pearson residual (8.86) is capped at 6 in the figure, following the package’s plotting convention.

5.4 Retrospective patterns

Nine retrospective peels were run from the Rceattle fit. The unpeeled model ends in 2024, and the peeled fits end in 2023 through 2015. Each peel preserves the data-availability pattern of the 2024 model. Thus, fishery age compositions end one year before the peel terminal year, survey series retain their historical sampling schedule and terminal lags, and earlier discontinued series retain their original ending years.

Each peel applies the fitting sequence documented in the appendix. The terminal-year fishery and mirrored catch-per-unit-effort selectivity increments are fixed at zero. All nine final fits fully met the convergence criteria, with maximum gradients from 2.1e-05 to 6.4e-05.

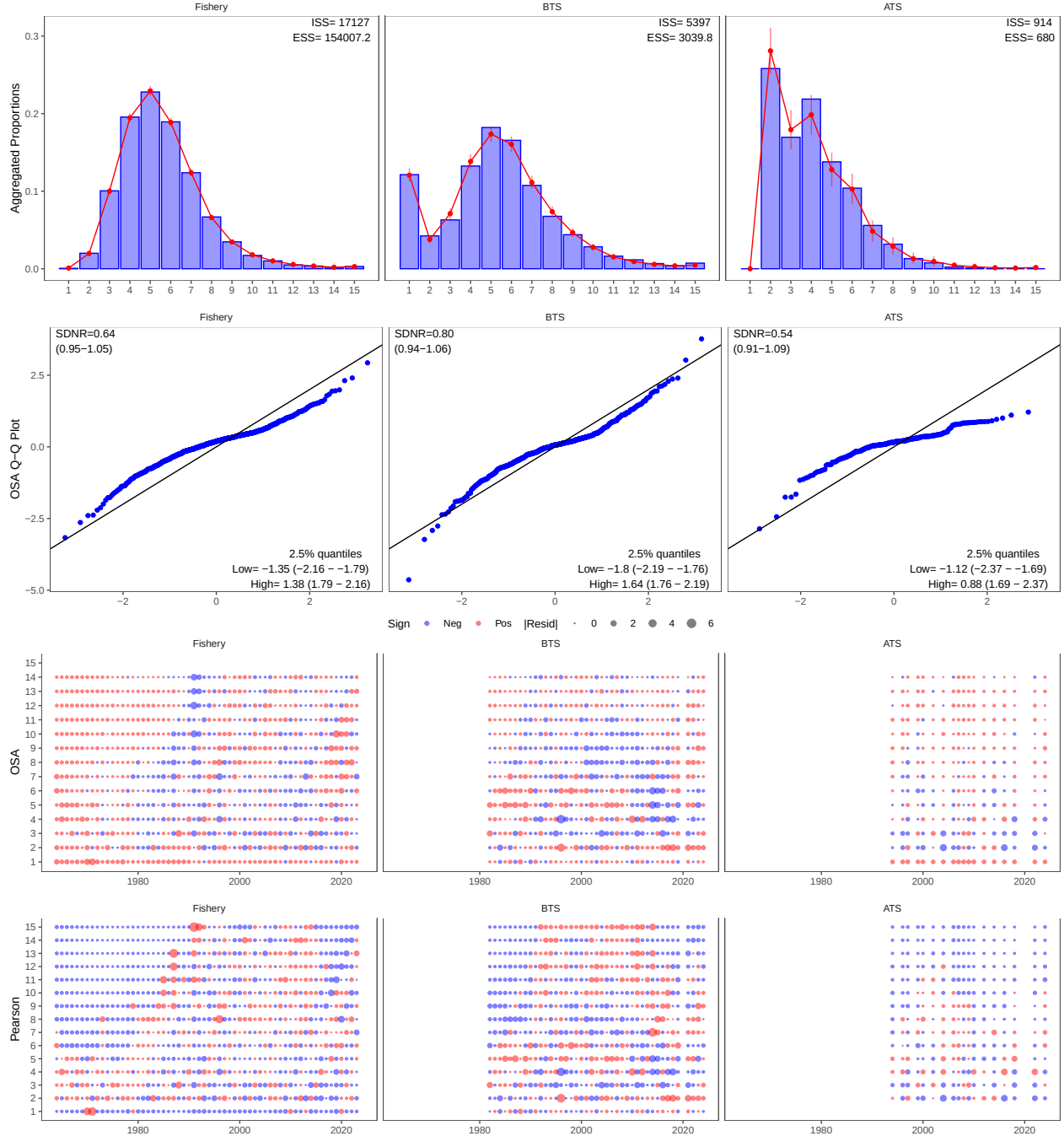


Figure 5: Age-composition diagnostics for the Rceattle fit. The fit uses complete ages 1–15, nominal fishery sample sizes, and BTS and ATS integer sample sizes aligned with the modified ADMB model. Diagnostics were produced with afscOSA for the fishery, bottom trawl survey (BTS), and acoustic-trawl survey (ATS). The top row compares aggregate observed proportions (blue bars) with fitted proportions (red points and lines); red vertical ranges contain 95% of data simulated conditionally from the fitted multinomial model, aggregate sample size, and estimated parameter values held constant. The second row shows quantile–quantile plots of Rceattle’s internal one-step-ahead (OSA) residuals with the standard deviation of normalized residuals (SDNR), its expected 95% interval, and lower and upper tail diagnostics. The lower rows show OSA and Pearson residuals by age and year; red and blue indicate positive and negative residuals, and bubble area represents magnitude. Absolute residuals greater than 6 are capped at 6 for display. The fitted model retains ATS 2020; its sample size of one yields zero rounded counts in the afscOSA plotting conversion, so the plotting layer omits that row.

The NonParametricPM fishery selectivity is parameterized at age. The base `sel_coff` parameters and annual `sel_coff_dev` random-walk increments apply to age bins; the fishery uses 12 estimated bins and holds older ages at the last estimated value. Fixing every terminal-year increment at zero therefore holds the realized terminal selectivity-at-age curve at the preceding-year curve. Across all peels and both the fishery and mirrored catch-per-unit-effort series, the maximum absolute terminal-to-preceding-year difference was $8.9\text{e-}16$. These Rceattle fits are separate from the legacy AD Model Builder (ADMB) retrospective files. The model runs with earlier terminal years generally estimate greater spawning biomass and recruitment and lower fishing mortality over their shared years (Figure 6).

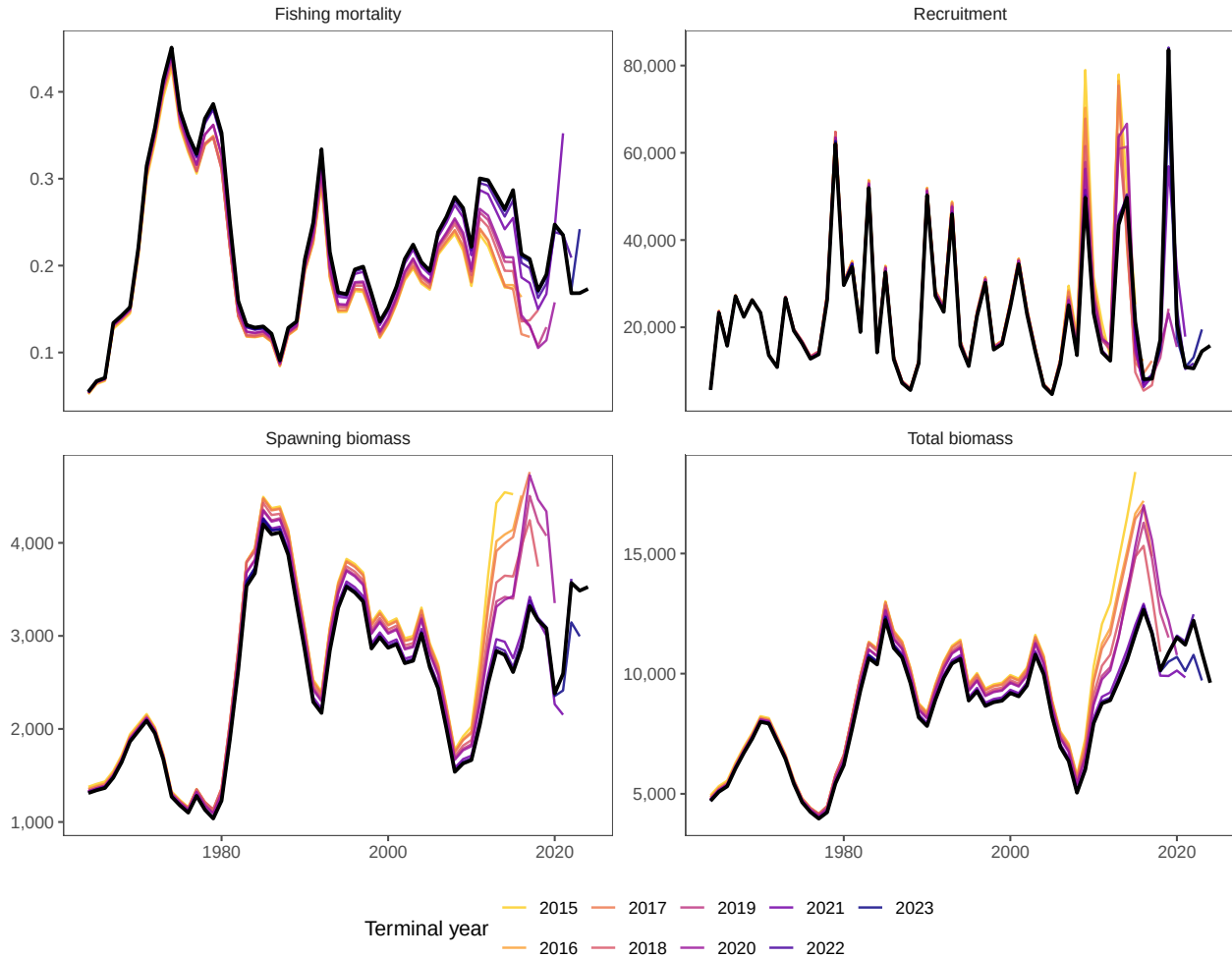


Figure 6: Nine retrospective peels for the Rceattle configuration. The black line is the unpeeled model ending in 2024; colored lines are successive peels ending in 2023 through 2015 and are plotted through each peel’s terminal year. Each peel preserves the original data-availability lags and fixes the terminal fishery/CPUE selectivity-at-age increment at zero, holding the terminal curve at its preceding-year value. All nine fits fully met the convergence criteria.

The estimated Mohn’s rho values are 0.261 for spawning biomass, 0.119 for total biomass, 0.147 for recruitment, and -0.077 for fishing mortality. Across the recent nine-year period, the available data and model tended to overestimate recruitment and spawning biomass when recent years of data were successively removed. Fishing mortality showed the opposite pattern, with lower estimates when recent data were removed. Spawning biomass shows the largest average relative difference. The fully met convergence criteria and small gradients support interpretation of these values as properties of the revised retrospective specification, while the magnitude of spawning-biomass rho motivates continued evaluation of selectivity, survey influence, and recent recruitment estimates (Table 5).

Table 5: Mohn’s rho for the nine retrospective peels of the Rceattle configuration. Values are the mean relative difference between a peeled estimate in its terminal year and the corresponding estimate from the unpeeled model. Each peel preserves the 2024 data-availability pattern and fixes the terminal fishery/CPUE selectivity-at-age increment at zero.

Quantity	Peels	Mohn’s rho
Total biomass	9	0.119
Spawning biomass	9	0.261
Recruitment	9	0.147
Fishing mortality	9	-0.077

5.4.1 Cohort revision and data-source diagnostics

The elevated spawning biomass in model runs with terminal years set to 2015–2020 is associated primarily with the initially large estimates of the 2013 and 2014 year classes, with a smaller contribution from the 2012 year class. Those cohorts support much of the estimated spawning biomass during 2015–2020 as they recruit to the mature population. The purpose of the following diagnostics is to learn which data signals and model processes cause their estimated strength, and consequently spawning biomass, to shift downward when later observations are included. For example, between model runs with terminal years set to 2020 and 2021, estimated age-1 abundance declines from 63,618 to 45,548 million fish for the 2013 cohort and from 66,666 to 49,473 million fish for the 2014 cohort. The corresponding estimate for the 2012 cohort declines from 15,947 to 12,622 million fish.

This uncertainty display is directly comparable in intent to the historical ADMB retrospective figure supplied for reference. Both show the same broad qualitative feature: earlier terminal-year fits produce higher spawning biomass over the mid-2010s, but with substantial uncertainty. The comparison does not establish identical causes because the ADMB and Rceattle retrospectives differ in implementation and fitted configuration. The Rceattle bands also represent approximate, conditional Hessian uncertainty; they should not be interpreted as uncertainty in the retrospective bias itself (Figure 7).

The transition from the 2020 to 2021 peel is especially informative because it contains the main downward step. Five records become available: the 2020 fishery age composition (effective sample size 565.2), the 2021 BTS biomass index (4,889 kt), the 2021 BTS age composition (effective sample size 149), the 2021 AVO index, and the BTS age-1 index record. Controlled refits began at the converged all-data 2021 solution and used the same parameter map. Each refit removed one newly available observation block, so these results measure conditional influence rather than providing independent alternative models.

The survey context makes this transition unusual. NOAA Fisheries canceled the 2020 eastern Bering Sea bottom-trawl survey, along with the standard vessel-based pollock acoustic survey, because the COVID-19 pandemic prevented safe survey operations (NOAA Fisheries 2020). The missing BTS observation therefore created a two-year interval between the 2019 and 2021 BTS biomass estimates. AFSC used three saildrone uncrewed surface vehicles to collect acoustic observations across the standard pollock survey area as a substitute for the research-vessel acoustic sampling. That effort provided useful abundance information, but it lacked the trawl samples used to identify species and estimate the age and size composition of the acoustic signal. NOAA consequently characterized the information as less detailed and more uncertain than a standard vessel survey, and the assessment assigned the estimate greater observation error (NOAA Fisheries 2020). This combination of a missing BTS year and a higher-uncertainty acoustic substitute reduces the continuity of fishery-independent information available to track the 2012–2014 cohorts before the 2021 BTS observation enters.

Removing the 2021 BTS biomass index increases the 2013 and 2014 recruitment estimates by 10.2% and 14.5%, respectively, and increases 2020 spawning biomass by 27.6%. Removing the 2020 fishery age composition increases the two cohort estimates by 19.5% and 10.8% and 2020 spawning biomass by 14.8%. In contrast, removing the 2021 BTS age composition changes these quantities by only about 0–4%, removing AVO

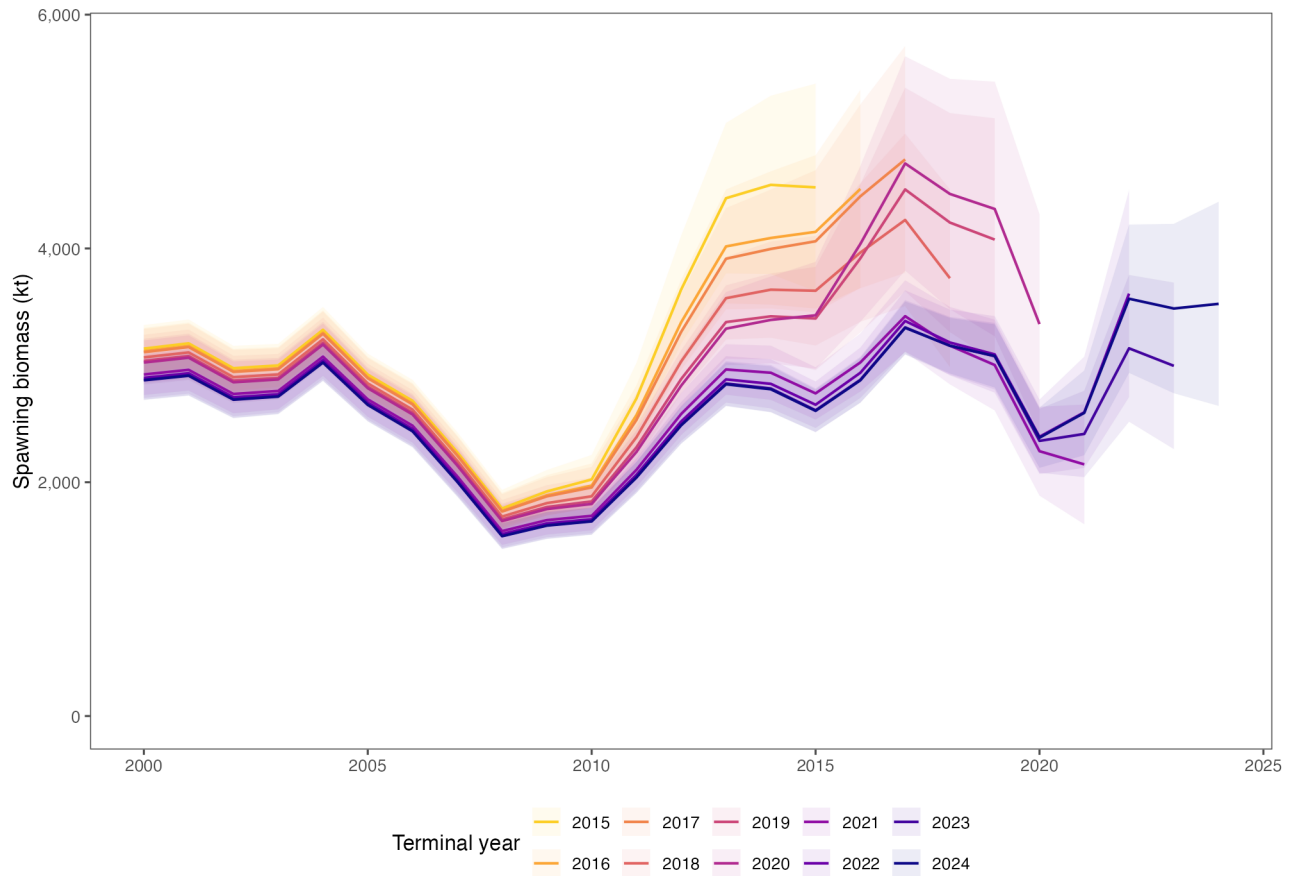


Figure 7: Rceattle retrospective spawning-biomass estimates with approximate plus or minus two-standard-deviation intervals from each peel's Hessian-based covariance calculation. The intervals describe within-peel estimation uncertainty conditional on each fitted model; they do not include structural uncertainty or covariance among peels.

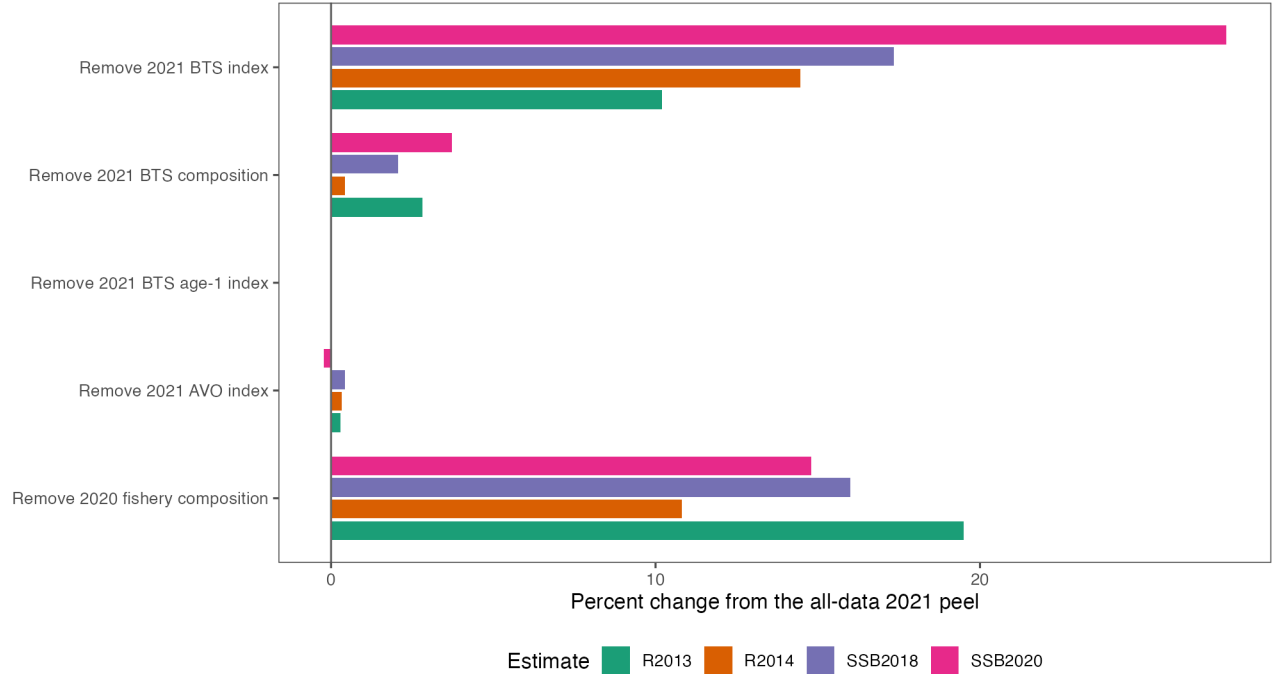


Figure 8: Influence of observation blocks entering between the 2020 and 2021 peels. Bars show the percentage change from the all-data 2021-peel estimate after removing one block and refitting from the converged 2021 solution with the same parameter map.

changes them by less than 0.5%, and removing the BTS age-1 index has no effect on these older cohorts. The cumulative addition experiment gives the same ordering: adding the 2020 fishery composition produces the first major decline, adding the 2021 BTS index produces the second, and the BTS composition and AVO index make smaller subsequent adjustments (Figure 8; Table 6).

The composition traces clarify why the fishery record is influential. For the 2013 cohort at age 8, the 2020 fishery observed proportion is 0.063 compared with a fitted proportion of 0.110. The 2021 BTS composition also places less mass on that cohort at age 9 (0.042 observed versus 0.070 fitted), but its conditional leverage is smaller. Evidence for the 2014 cohort is less uniform: its fishery and BTS terminal proportions are close to their fitted values. The 2013 cohort therefore shows a recognizable late-age composition signal across two data sources, whereas the 2014 downward revision is tied more strongly to the overall BTS abundance scale and the model’s allocation of that abundance among ages (Figure 9).

Selectivity is an important mediator rather than a separate observation. When the 2021 peel is refit with BTS selectivity fixed at its realized 2020-peel curves, the 2013 and 2014 recruitment estimates increase by 14.5%

Table 6: Controlled cumulative addition of observations entering the 2021 peel. Percentages are relative to the complete 2021-peel fit. All fits fully met the convergence criteria.

Scenario	R2013 (%)	R2014 (%)	SSB 2018 (%)	SSB 2020 (%)	Maximum gradient
2020 data availability	39.8	29.5	38.9	44.7	4.68 × 10 ⁻³
Add 2020 fishery composition	10.0	10.5	14.1	25.1	1.07 × 10 ⁻³
Add 2021 BTS index	3.1	0.7	2.4	3.6	4.28 × 10 ⁻⁴
Add 2021 BTS composition	0.3	0.3	0.4	-0.2	4.59 × 10 ⁻⁵
Add 2021 AVO index	0.0	0.0	0.0	0.0	4.43 × 10 ⁻⁶

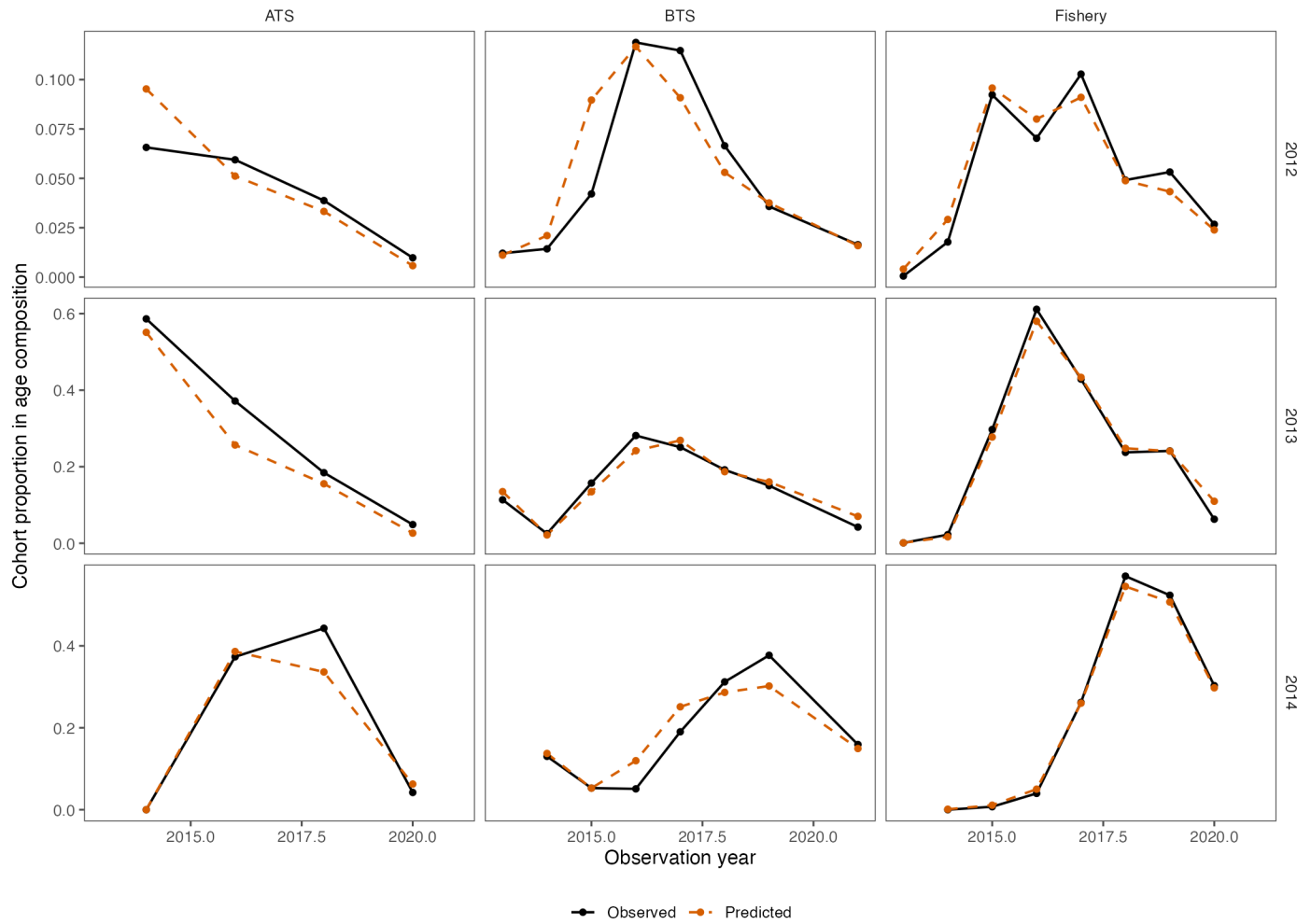


Figure 9: Observed and fitted contributions of the 2012–2014 cohorts to ATS, BTS, and fishery age compositions through the 2021 peel. Each point follows a cohort as it ages; differences are proportions within an annual age composition, not survival estimates.

Table 7: Selectivity-control refits of the 2021 peel. Each listed fleet’s selectivity parameter fields were fixed at the 2020-peel estimates; percentages are relative to the complete 2021-peel fit. These are diagnostic controls, not candidate assessment configurations.

Scenario	R2013 (%)	R2014 (%)	SSB 2018 (%)	SSB 2020 (%)	Maxim
All selectivity re-estimated	0.0	0.0	0.0	0.0	
Fishery and CPUE fixed at 2020 peel	3.1	-4.0	-4.9	-4.7	
BTS fixed at 2020 peel	14.5	14.9	19.9	19.2	

and 14.9%, and 2020 spawning biomass increases by 19.2%, relative to the all-data 2021 fit. Fixing fishery and CPUE selectivity at their 2020-peel curves has much smaller and mixed effects. Thus, the lower 2021 BTS index supplies the main new abundance-scale information, and re-estimated BTS selectivity determines how much of that scale change is assigned to these cohorts (Table 7).

These experiments narrow the source of the shift but do not yet identify a unique biological process. Natural mortality is fixed in this configuration, so the model cannot represent cohort-specific anomalous survival directly. The same observations could instead reflect changing survey availability or movement, ageing or composition inconsistencies, or selectivity changes that are not fully separated from abundance. The next diagnostic stage should therefore (1) audit the age determinations and sample construction for the 2013 cohort in the 2020 fishery and 2021 BTS samples, (2) map the spatial and size composition of the 2019 and 2021 BTS observations to test availability changes across the survey gap, (3) profile or otherwise perturb BTS selectivity while retaining the 2021 index, and (4) run a targeted cohort-survival or time-varying-mortality sensitivity. Until those checks are complete, the evidence supports a data-conflict/selectivity explanation for the downward revision, not a definitive attribution to mortality or any single survey record.

6 Projections

The projections carry the terminal Rceattle population state and biological schedules forward under the seven Tier 3 harvest alternatives implemented in the Standard Projection Model (J. Ianelli 2026). Fishery selectivity and fishing mortality represent average conditions during 2020–2024. All alternatives use the same assessment results and fixed catches for 2025 and 2026, after which their harvest rules produce different catch, biomass, and fishing-mortality trajectories. The technical conversion, execution steps, and validation checks are documented in Appendix A-2.

Table 8: Projection files generated from the saved Rceattle fit for the spmR/SPM calculations.

spmR Projection Files

file	role	exists
spm.dat	SPM setup	TRUE
pm.prj	Rceattle-derived species input	TRUE
tacpar.dat	TAC parameters	TRUE
spm	SPM executable	TRUE
age_schedules.csv	Rceattle-derived age schedules	TRUE
fishery_selectivity_recent_years.csv	Annual fishery selectivity used in the recent-year average	TRUE

The saved projection products include the population inputs, projection controls, simulation output, and summary tables used in this section (Table 8).

Table 9: Key settings and source information for the Rceattle-derived SPM projection.

Generated SPM Setup and Source Information

Item	Value
Rceattle source fit	nonparametric_pm
Composition likelihood	Multinomial with $c = 0.001$
Composition sample sizes	fishery nominal; BTS and ATS integer sample sizes aligned with modified ADMB
spmR version	0.3.0
Terminal assessment year	2024
Fishery selectivity rule	arithmetic mean of annual selectivity-at-age
Fishery selectivity years	2020, 2021, 2022, 2023, 2024
Recent average-F years	2020, 2021, 2022, 2023, 2024
Spawning timing	0.25 year from January 1 (start of April)
Rceattle encoding	spawn_month = 3 elapsed months; $3/12 = 0.25$
SPM encoding	Spawnmo = 4 as a one-based calendar-month code; $(4 - 1)/12 = 0.25$
Timing compatibility check	Matched: both calculations apply survival through 0.25 of the year
Spawning-biomass timing convention	Mature spawning biomass after survival through 0.25 of the year: $N \times \exp(-0.25 Z) \times \text{maturity} \times \text{spawning weight}$
Projection begin year	2025
Projection years	14
Simulations	1000
Alternatives	1, 2, 3, 4, 5, 6, 7

The projection setup identifies the assessment source, biological timing, averaging years, simulation count, and harvest alternatives (Table 9).

Table 10: SPM projection alternatives included in the generated setup file.

SPM Projection Alternatives

Alternative	Name	Description
1	Maximum permissible ABC	Maximum permissible Tier 3 ABC harvest rate.
2	Author-specified ABC	Author-specified adjustment to the Tier 3 ABC harvest rate; the current multiplier is 1.0.
3	Average recent F	Recent average fishing mortality from the Rceattle-derived input file.
4	Alternative SPR rate	User-specified SPR rate of 0.60.
5	No fishing	Zero fishing mortality after the fixed-catch years.
6	OFL threshold determination	OFL harvest rate for threshold and status-determination calculations.
7	Status-determination ramp	Maximum permissible ABC for three years followed by the OFL status-determination calculation.

The seven alternatives span maximum permissible ABC, recent fishing mortality, a higher spawning-potential-ratio target, no fishing, and status-determination calculations (Table 10).

6.1 Seven Tier 3 Scenario Results

Catches for 2025 and 2026 are fixed at 1,350 thousand t in every alternative; the alternatives therefore first diverge in 2027 (Table 11). The reported values are simulation means from 1,000 SPM simulations.

Table 11: Tier 3 projection results for all seven FMP alternatives using inputs generated from the Rceattle EBS pollock model. Catch and biomass quantities are in thousand t; $B/B_{35\%}$ is spawning biomass relative to the Tier 3 $B_{35\%}$ proxy.

Seven Tier 3 Projection Alternatives

Alt	Scenario	Year	Catch	ABC	OFL	SSB	F	B/B35
1	Maximum permissible ABC	2027	1,864	1,864	2,259	2,476	0.350	1.17
1	Maximum permissible ABC	2028	1,308	1,308	1,583	2,115	0.310	1.00
2	Author-specified ABC	2027	1,864	1,864	2,259	2,476	0.350	1.17
2	Author-specified ABC	2028	1,308	1,308	1,583	2,115	0.310	1.00
3	Average recent F	2027	1,132	1,132	2,259	2,582	0.200	1.22
3	Average recent F	2028	1,015	1,015	1,963	2,446	0.200	1.16
4	Alternative SPR rate	2027	856	856	2,259	2,619	0.150	1.24
4	Alternative SPR rate	2028	803	803	2,103	2,585	0.150	1.22
5	No fishing	2027	0	0	2,259	2,726	0.000	1.29
5	No fishing	2028	0	0	2,455	3,034	0.000	1.44
6	OFL threshold determination	2027	1,561	1,561	1,561	2,017	0.370	0.95
6	OFL threshold determination	2028	1,210	1,210	1,210	1,831	0.340	0.87
7	Status-determination ramp	2027	1,509	1,809	1,809	2,211	0.320	1.05
7	Status-determination ramp	2028	1,424	1,424	1,424	1,977	0.360	0.94

Table 13: Candidate ABC projection given assumed future catches. Catch is fixed at 1,300 thousand t; ABC is the maximum permissible ABC calculated by SPM. $B/B_{35\%}$ is mean spawning biomass relative to the Tier 3 proxy.

Candidate ABC Projection Given Assumed Future Catches

Year	Catch	ABC	OFL	Mean B	$B/B_{35\%}$
2025	1,300	2,360	2,878	3,523	167%
2026	1,300	2,131	2,588	3,013	143%
2027	1,300	1,891	2,295	2,597	123%
2028	1,300	1,583	1,909	2,366	112%
2029	1,300	1,446	1,763	2,364	112%
2030	1,300	1,459	1,786	2,446	116%
2031	1,300	1,520	1,871	2,547	120%
2032	1,300	1,594	1,967	2,644	125%

Table 12: Age-specific schedules applied in SPM. Natural mortality, maturity, and weights come from the terminal Rceattle configuration; fishery selectivity is the arithmetic mean at age over 2020–2024.

Projection Age-Specific Schedules

Age	Spawning_weight	Fishery_weight	Maturity	Natural_mortality	Selectivity
1	0.085	0.025	0.000	0.900	0.000
2	0.196	0.186	0.008	0.450	0.061
3	0.322	0.401	0.289	0.300	0.220
4	0.534	0.667	0.641	0.300	0.414
5	0.621	0.663	0.842	0.300	0.775
6	0.707	0.720	0.901	0.300	1.031
7	0.822	0.791	0.947	0.300	1.302
8	0.946	0.905	0.963	0.300	1.530
9	1.051	1.009	0.970	0.300	1.456
10	1.113	1.061	1.000	0.300	1.394
11	1.178	1.138	1.000	0.300	1.406
12	1.290	1.257	1.000	0.300	1.353
13	1.332	1.267	1.000	0.300	1.353
14	1.367	1.290	1.000	0.300	1.353
15	1.415	1.368	1.000	0.300	1.353

The projection retains the terminal natural mortality, maturity, and weight schedules and uses mean fishery selectivity at age for 2020–2024 (Table 12).

6.2 Alternative 2 Fixed-Catch Projection

This additional run applies Alternative 2 and fixes catch at 1,300 thousand t for 2025–2032. The projection horizon includes every fixed-catch year, and spawning biomass remains above the Tier 3 $B_{35\%}$ proxy throughout the reported period (Table 13).

6.3 Projection Figures

Projected spawning biomass remains above $B_{35\%}$ under the harvest alternatives over the displayed horizon, with trajectories diverging after the fixed-catch years. Lines show simulation medians and ribbons show central 90% intervals (Figure 10).

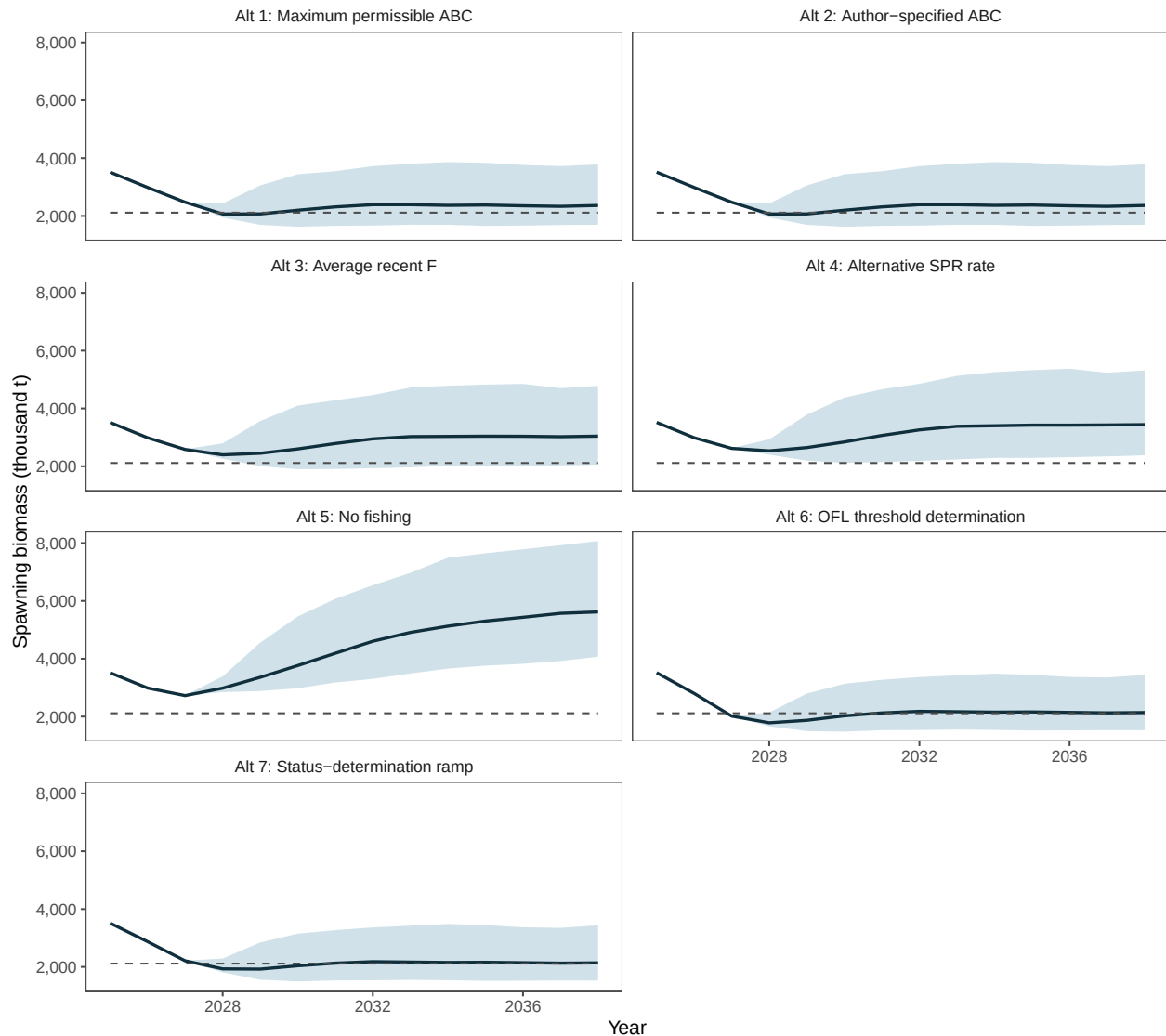


Figure 10: Projected spawning biomass under all seven Tier 3 alternatives. Lines are simulation medians, ribbons span the 5th to 95th percentiles, and dashed lines mark $B_{35\%}$.

Projected catch, ABC, and OFL are identical during the fixed-catch period and then separate according to the alternative harvest rules (Figure 11).

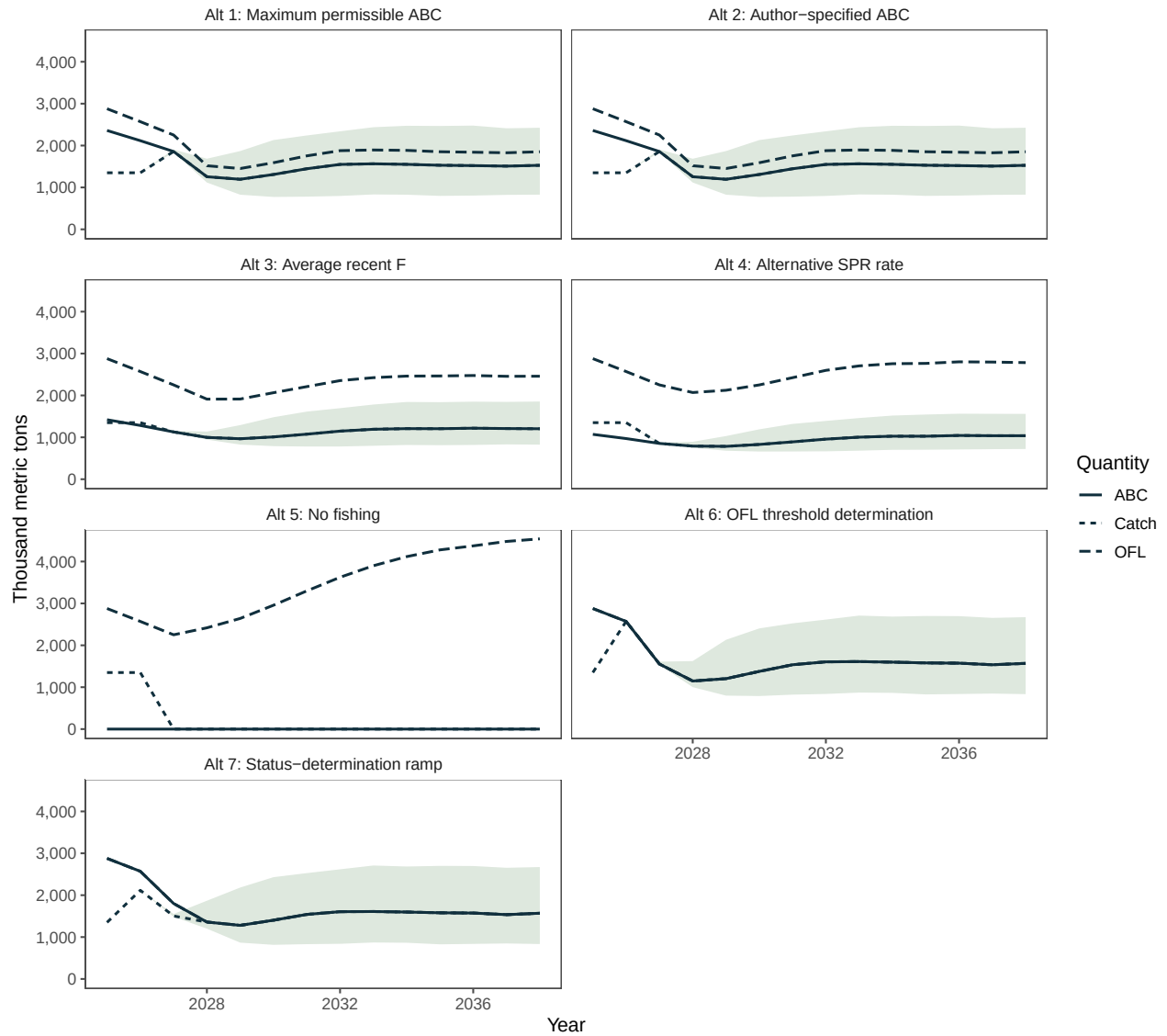


Figure 11: Projected catch, ABC, and OFL under all seven Tier 3 alternatives. Lines are simulation medians and catch ribbons span the 5th to 95th percentiles.

7 Exploratory DSEM Recruitment Pathways

This section is the indicator-analysis pilot within a draft EBS pollock Ecosystem and Socioeconomic Profile (ESP), rather than an attempt to estimate the full pollock early-survival system. The BSAI Groundfish Plan Team recommended in 2021 that an EBS walleye pollock ESP be added to the list of future ESPs ([Bering Sea/Aleutian Islands Groundfish Plan Team 2021](#)). The analysis follows the sequence recommended in the AFSC causal-modeling guidance: define the outcome, build a literature-based mechanism table and generic DAG, select a small set of scientifically defensible graphs, retain a no-edge baseline, and compare all graphs on the same data ([Monnahan and Shotwell 2026](#)). The mechanism table and DAG draw on the January 2025 pollock workshop, the age-0 pollock working-group material, and the Tiger Team data-availability synthesis ([Monnahan et al. 2026](#); [Siddon et al. 2026](#); [Alaska Fisheries Science Center Age-0 Pollock Causal-Modeling Working Group 2026](#); [EBS Pollock Tiger Team 2026](#)) and the [EBS pollock Early Survival Pathways synthesis](#) ([J. N. Ianelli 2026](#)).

The response is the Rceattle age-1 recruitment state. The two examples test late-summer SST in the cohort’s age-0 year and cold-pool extent in the following age-1 calendar year. They were selected because both have relatively long annual records and represent distinct, prespecified hypotheses. The analysis does not treat them as direct measurements of prey quality, energetic condition, cannibalism, or survival.

7.1 ESP development scope

The September product documents the justification, synthesis, generic DAG, mechanism table, and two illustrative fits. It does not substitute a long list of available covariates for a small set of causal hypotheses. The intended progression identifies the completed September analyses and the work deferred to November (Table 14). The full draft ESP structure is provided in Appendix A-3.

Table 14: Development scope for the EBS pollock ESP and its Rceattle-DSEM indicator analysis.

Component	September draft	November target
Justification and outcome	Document the Plan Team request and define age-1 recruitment as the focal population response.	Confirm the focal response and management use with the ESP team.
Indicator synthesis	Provide a life-history synthesis, generic DAG, mechanism table, and data-availability review.	Update indicators through the ESP and Ecosystem Status Report data-collection process.
Candidate hypotheses	Retain two transparent proxy examples: age-0 late-summer SST and cold-pool extent.	Fit two to five defensible candidate DAGs that represent proximal mediators where data permit.
Indicator analysis	Compare the two single-edge examples and their combined diagnostic with the no-edge IID reference.	Add predictive, retrospective, timing, and observation-error checks to the candidate-DAG comparison.
Management use	Use results to illustrate the integrated Rceattle-DSEM workflow and identify data gaps.	Determine whether validated indicators are suitable for operational recruitment modeling or risk-table context.

7.2 Mechanism table and generic DAG

The reduced mechanism table distinguishes processes that can be tested with the current annual data from biologically important mediators that remain intermittent or unavailable (Table 15). Expected signs describe the direction from the listed driver toward age-1 recruitment, whether directly or through the stated mechanism.

Table 15: Reduced mechanism table for the illustrative age-0 to age-1 recruitment analysis. Coverage refers to the cohort-aligned data presently available to this working paper.

Driver	Mechanism	Expected sign	Current coverage	Role here
Sea ice	Winter ice affects summer cold-pool extent, bloom conditions, and prey production.	Indirect	Annual physical index available; not fitted here	Upstream context
Cold-pool extent	A larger cold pool can separate juveniles from adult pollock and reduce cannibalism.	Positive	42 annual values	Illustrative proxy edge
Age-0 late-summer SST	Warmer water can increase metabolic demand and reduce energy stored before winter.	Negative	41 annual values	Illustrative proxy edge
Large copepods and euphausiids	Lipid-rich prey supports age-0 growth, energy storage, and overwinter survival.	Positive	Intermittent; 15 values in the current ESP cohort table	Unfitted mediator
Fall age-0 energy	Higher energy density and total energy increase overwinter survival to age 1.	Positive	Absent from the current annual cohort table	Priority mediator
Adult-pollock overlap	Greater adult-juvenile overlap increases cannibalism exposure.	Negative	A separate range-overlap product exists, but its survey timing is contemporaneous with the age-1 state	Deferred from the main example

The generic DAG is adapted from the age-0 pollock working-group graph and retains both bottom-up and top-down routes (Figure 12). It makes the unobserved or intermittent mediators visible instead of allowing the two available physical proxies to stand in for entire mechanisms. Solid arrows are conceptual. Dashed arrows are the only recruitment paths fitted in this proof of concept.

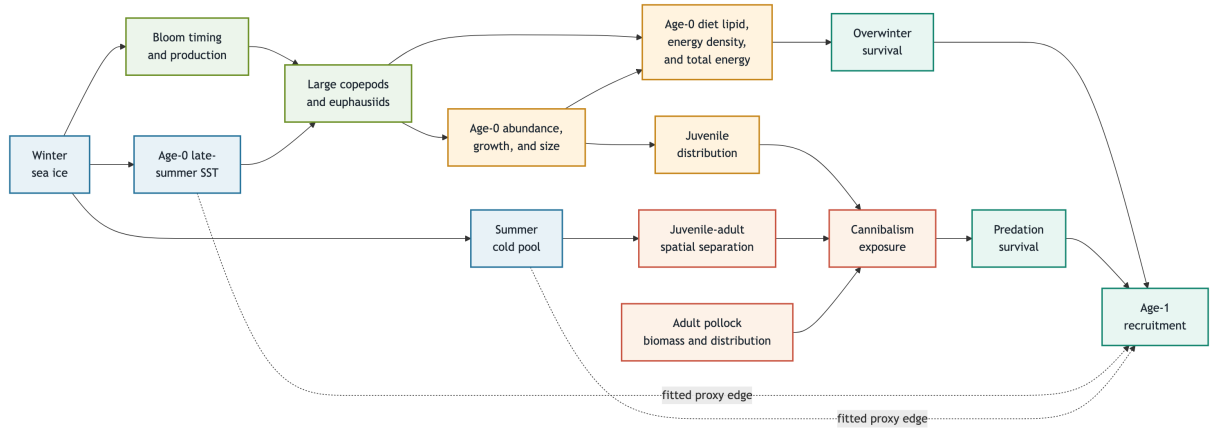


Figure 12: Generic directed acyclic graph for eastern Bering Sea pollock survival from age 0 to age 1. Solid arrows summarize hypothesized mechanisms. Dashed arrows are the two direct proxy paths fitted in the illustrative Rceattle-DSEM comparison.

The generic graph is intentionally broader than the statistical examples. It shows that a negative SST coefficient would be consistent with an energetic constraint, but would not identify prey quality, age-0 energy, or overwinter survival as the active mediator. Likewise, a cold-pool coefficient would be a proxy for spatial separation and cannibalism unless overlap and mortality were observed directly.

7.3 Two illustrative hypotheses

The proof of concept compares two single-edge hypotheses with a common reference:

- **Energetic-condition proxy:** higher age-0 late-summer SST is associated with lower age-1 recruitment because warmer conditions can increase metabolic demand and reduce fall energy reserves (Heintz et al. 2013; Siddon et al. 2013; Duffy-Anderson et al. 2016).
- **Spatial-separation proxy:** larger cold-pool extent is associated with higher age-1 recruitment because it can reduce juvenile-adult spatial overlap and cannibalism.

The age-0 SST value is carried forward to the DSEM row for that cohort's age-1 recruitment. The zero-lag path is therefore cohort-aligned rather than a claim that SST was measured in the age-1 year. In this

section “IID” refers to the no-recruitment-edge reference model. The combined model is retained as a limited diagnostic of whether the two proxy edges add information when fitted together; it is not a third mechanistic hypothesis.

All four specifications use the same observation-year universe, missingness pattern, standardization, and AR(1) processes for the observed covariates. Within this common data block, AIC compares relative fit after accounting for model complexity. The percentage reduction in residual recruitment variation compares each model’s estimated unexplained recruitment variance with the IID reference. It is an in-sample description, not a causal R-squared or a predictive-skill score.

7.4 Illustrative results

Table 16: Illustrative Rceattle-DSEM model comparison. Delta AIC and the percentage reduction in residual recruitment variation are calculated within the common observation block.

Model	AIC	Delta AIC	Residual variation reduced (%)
Age-0 late-summer SST	4,144.56	0.00	10.5
Cold pool + age-0 late-summer SST	4,146.19	1.63	12.0
IID recruitment deviations	4,146.81	2.25	0.0
Cold pool	4,148.66	4.11	0.3

All four fits fully met the convergence criteria, had positive-definite Hessians, and had maximum absolute gradients no larger than $9.4\text{e-}05$. The age-0 late-summer SST model had the lowest AIC and reduced estimated residual recruitment variation by 10.5% relative to IID. The cold-pool-only model reduced residual variation by 0.3% and had higher AIC than IID. The combined model reduced residual variation by 12.0%, but its additional parameter left its AIC above the SST-only model (Table 16; Figure 13).

Table 17: Standardized proxy paths to age-1 recruitment. Intervals are the estimate plus or minus 1.96 standard errors.

Covariate	Estimate	Standard error	Lower 95%	Upper 95%
Cold pool				
Cold pool	0.041	0.107	-0.169	0.251
Age-0 late-summer SST				
Age-0 late-summer SST	-0.222	0.104	-0.426	-0.017
Cold pool + age-0 late-summer SST				
Cold pool	-0.069	0.114	-0.292	0.153
Age-0 late-summer SST	-0.254	0.118	-0.485	-0.024

The SST coefficient was negative and its interval excluded zero in both the single-edge and combined models. The cold-pool coefficient was small and its interval included zero in both models. In plain language,

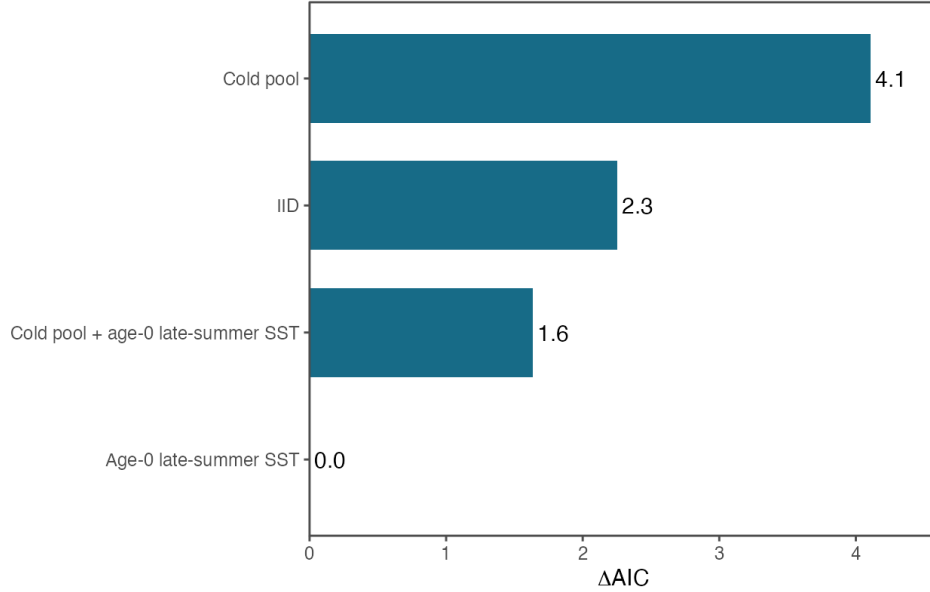


Figure 13: Akaike information criterion differences for the illustrative DSEM models. The age-0 late-summer SST model has the lowest AIC.

the current annual data contain a modest in-sample association consistent with the warm-year energetic-constraint hypothesis, while the direct cold-pool proxy edge received little support. Neither result establishes the intervening mechanism (Table 17; Figure 14).

The missingness-preserving randomization check replaced both environmental series with independent standard-normal values using seed 20260812. None of the randomized recruitment paths improved on the randomized IID model: their delta AIC values ranged from 1.03 to 2.95, and every randomized path interval included zero. This is a basic check against an identical signal arising from arbitrary series with the same gaps; it does not replace out-of-sample validation.

7.5 Interpretation and next test

This proof of concept supports keeping the candidate set small. The SST result is useful as an illustration of the Rceattle-DSEM connection, but SST is an upstream proxy. The next scientifically stronger test is to insert a proximal mediator from the mechanism table: fall age-0 total energy or energy density, followed by a lipid-rich large-copepod/euphausiid index when enough cohort years are available (Eisner et al. 2020; Heintz et al. 2013; Siddon et al. 2013). That sequence would test whether the SST association weakens when the hypothesized energetic path is represented directly.

A top-down overlap example should return only after its timing and uncertainty are aligned with the age-0-to-age-1 survival interval. The available adult-pollock overlap product is informative, but its survey occurs after the start-of-year age-1 state and it is partly population-derived. Including it here would blur a causal example with a contemporaneous association.

Before either proxy is considered for operational recruitment inference, the paths should be tested with terminal-year or blocked prediction, retrospective stability, alternative cohort timing, and observation-error sensitivity. These DSEM results remain separate from the assessment configuration and projection results.

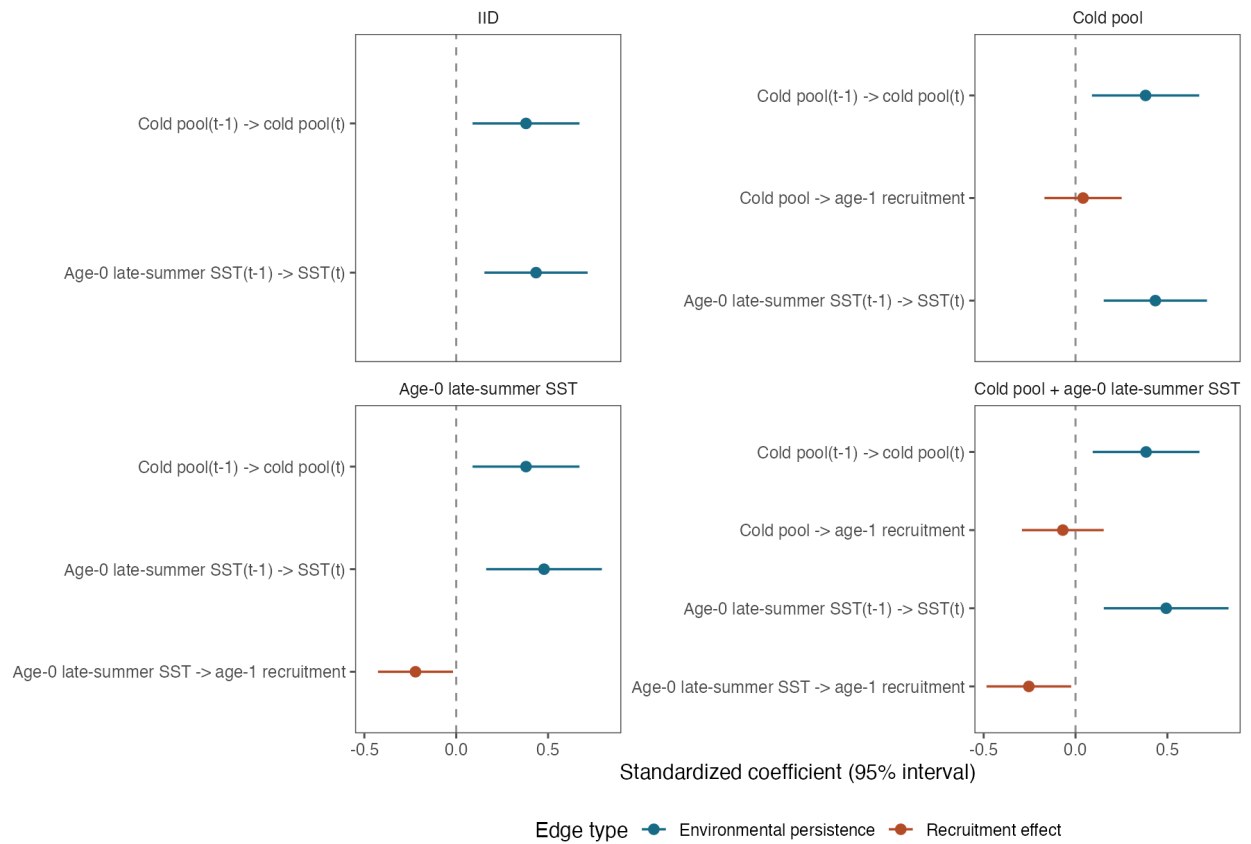


Figure 14: Standardized coefficients for the illustrative DSEM models. Points are estimates and horizontal lines are 95% intervals. Blue edges describe temporal persistence in observed covariates; orange edges are the two direct proxy paths to age-1 recruitment.

8 Summary and conclusions

When Rceattle parameters and fishery selectivity are set to the values used in the ADMB model, Rceattle reproduces spawning biomass and predicted catch to numerical precision. This establishes agreement in the tested population-dynamics calculations and parameter mapping. When Rceattle is fit to the data, the trajectories differ more, and the residual least-identified direction remains in the shared ATS/AVO annual selectivity deviations. The added ATS/AVO coefficient-increment penalty reduced the condition number below the convergence-warning threshold.

The diagnostic results identify a focused development path for the September 2026 Plan Team evaluation. The Rceattle fit fully met the convergence criteria. Composition OSA residuals remain underdispersed. Nine terminal-year-consistent peels converge successfully and produce positive retrospective bias in spawning biomass, total biomass, and recruitment, together with negative bias in fishing mortality. The historical 2D AR1 sensitivity did not converge and retains its research status. The 50-jitter diagnostic recovered the best solution 48 times and found a poorer mode twice.

The scaled-back DSEM proof of concept compared two prespecified proxy edges with the no-edge IID reference. The age-0 late-summer SST model had the lowest AIC and reduced estimated residual recruitment variation by 10.5% relative to IID. The cold-pool edge received little support. This result illustrates the Rceattle-DSEM connection, but the annual data do not yet represent the prey, age-0 energy, overwinter-survival, and cannibalism mediators required to attribute the association to a causal mechanism.

Immediate work should retain the ADMB-to-Rceattle comparison as a repeatable numerical check, evaluate selectivity and survey influence on the retrospective pattern, and continue sensitivity checks for the shared ATS/AVO annual selectivity deviations. Given acceptance in other applications and the proven bridging aspects, we feel this approach should be adopted for operational management advice for November/December 2026. Future refinements will explore trade-offs in fitting data in alternative ways (e.g., using random effects approaches for selectivity variability, selecting alternatives to the robust-multinomial approach used in this example). The exploratory DSEM work provides a reproducible approach for adding cohort-aligned covariates, while remaining separate from the final assessment configuration and projection results.

Rceattle is broadly used beyond this EBS pollock application. The framework has been applied to single- and multispecies groundfish cases in the eastern Bering Sea and Gulf of Alaska, including configurations for pollock, Pacific cod, and arrowtooth flounder. Its shared data structure accommodates one or more stocks, one- or two-sex population dynamics, and multiple fisheries and surveys. The present bridge therefore represents an application of a broader assessment framework rather than a model developed only for EBS pollock.

Rceattle is also under active development. Continued work on its TMB model, data interfaces, diagnostics, simulation tools, and documentation is expanding the set of configurations that can be represented and tested. This development is an advantage for future assessment work, but it also reinforces the need to record package versions, preserve model inputs and fitted outputs, and repeat the ADMB bridge and convergence checks when consequential components change.

Only part of the available Rceattle feature set was explored here. This study focused on a single-species age-structured configuration, ADMB parameter mapping, selectivity, composition fitting, standard diagnostics, projections, and an exploratory DSEM extension. It did not fully evaluate joint multispecies estimation, bioenergetics-based predation mortality, estimable growth, environmental linkages across other biological parameters, alternative harvest control rules and climate scenarios, or closed-loop management strategy evaluation. These capabilities provide a substantial agenda for future testing, but each should be evaluated through controlled sensitivities and diagnostics before contributing to operational EBS pollock advice.

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Appendices

A-1 Reproducibility and analysis sequence

The working paper is rendered from saved model and diagnostic products so that building the document does not repeat long model fits. The commands below are run from the project directory, in the order shown when an earlier product is required by a later analysis.

1. The [input-preparation script](#) documents the transformations used to prepare the aligned Rceattle workbook; its precursor workbook is regenerated with Rscript "2024 EBS pollock setup data.R". The frozen workbook used for the reported fit is Data/EBS_24_pollock_m23_rceattle_full_1964-2024.xlsx. The same script documents the modified ADMB reference rebuild, run as `cd ADMB/m23_rceattle_full && admb pm && ./pm -nox -iprint 150` after the ADMB executable directory has been added to PATH.
2. The [two-stage fitting script](#) fits the reported Rceattle configuration and the 2D age-by-year AR1 sensitivity with Rscript "2024 EBS pollock.R". The [5.8.1 summary script](#) then generates the fitted-model summaries and standard-multinomial sensitivity using Rscript R/finalize_rceattle_5_8_1_analysis.R.
3. The [ADMB bridge runner](#) sets the Rceattle parameters and realized selectivity to the ADMB values and saves the trajectory, likelihood, and gradient comparisons with Rscript R/run_canonical_pm_forward_bridge.R.
4. The [OSA runner](#) generates the saved OSA residuals with Rscript R/run_canonical_pm_diagnostics.R. The [retrospective script](#) runs the nine terminal-year-consistent peels with Rscript "2024 EBS pollock two-stage retrospectives.R".
5. The [5.8.1 self-test runner](#) generates 50 phased simulations with Rscript R/run_rceattle_5_8_1_self_test.R. The [restart runner](#) repeats the same simulations from the estimated solution with Rscript R/run_rceattle_5_8_1_self_test_restart.R, and the [validation summarizer](#) is run with Rscript R/summarize_rceattle_5_8_1_validation.R.
6. The [projection runner](#), which uses the [Rceattle-to-SPM input writer](#), generates the seven-scenario and fixed-catch projection products with Rscript R/run_spmr_tier3_projection.R.
7. The [DSEM script](#) runs the four observed-covariate models with Rscript "2024 EBS pollock DSEM proof of concept.R" and the missingness-preserving randomized controls with Rscript "2024 EBS pollock DSEM proof of concept.R" --randomized.
8. The [compatibility-validation runner](#) and [simulation/projection checks](#) validate the pinned 5.20.0 source revision. The input-preparation script uses the canonical 5.20.0 workbook schema, and the controlled refit confirms that the regenerated workbook preserves the accepted fit.
9. After these saved products are available, the [report-input preparation script](#) extracts the compact quantities, diagnostics, and trajectories needed by the document with Rscript R/prepare_report_inputs.R. The HTML and PDF are then generated together with `scripts/render_report.sh`.

The report inventory identifies the model inputs, saved diagnostic products, scripts, and projection outputs needed to reproduce the working paper (Table 18).

R version 4.6.1 (2026-06-24)
Platform: aarch64-apple-darwin23
Running under: macOS Tahoe 26.6.2

Table 18: Files used or referenced by this working paper.

Role	Location	Available	Size
Prepared model-input workbook	Local workbook	TRUE	135.1 KB
BTS covariance input	Local data file	TRUE	12.7 KB
Compact report-ready Rceattle products	Versioned R serialized-data (RDS) file	TRUE	131.5 KB
OSA residuals	Local RDS file	TRUE	91.2 KB
OSA diagnostics	Local RDS file	TRUE	1.1 KB
Compact nine-peel retrospective products	Versioned RDS file	TRUE	21.1 KB
Nine-peel Mohn’s rho summary	Local CSV file	TRUE	0.2 KB
Nine-peel retrospective script	Local R script	TRUE	9.7 KB
ADMB-to-Rceattle comparison output	Local RDS file	TRUE	650.8 KB
Rceattle 5.8.1 self-test summary	Local CSV file	TRUE	0.5 KB
Rceattle 5.8.1 truncation diagnostics	Local CSV file	TRUE	0.4 KB
Rceattle 5.8.1 restart comparison	Local CSV file	TRUE	3.8 KB
Rceattle 5.8.1 jitter summary	Local CSV file	TRUE	2.3 KB
Rceattle 5.20.0 compatibility comparison	Versioned CSV file	TRUE	0.6 KB
ADMB-to-Rceattle comparison summary	Local comma-separated-values (CSV) file	TRUE	1.0 KB
Rceattle fitting script	Local R script	TRUE	15.7 KB
Diagnostics script	Local R script	TRUE	4.9 KB
Next-assessment update script	Local R script	TRUE	2.6 KB
ADMB comparison script	Local R script	TRUE	10.1 KB
Rceattle-to-SPM input writer	Local R script	TRUE	8.7 KB
spmR Tier 3 projection runner	Local R script	TRUE	1.5 KB
SPM projection manifest	Local CSV file	TRUE	0.5 KB
Exploratory DSEM script	Local R script	TRUE	17.1 KB
DSEM model comparison	Local CSV file	TRUE	0.5 KB
DSEM recruitment-path estimates	Local CSV file	TRUE	3.1 KB
Randomized DSEM control	Local CSV file	TRUE	0.5 KB
DSEM AIC comparison figure	Local PNG file	TRUE	24.3 KB
DSEM edge-coefficient figure	Local PNG file	TRUE	100.9 KB
Compact report-input manifest	Versioned CSV file	TRUE	6.9 KB


```

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib;  LAPACK version 3.

locale:
[1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8

time zone: Europe/Brussels
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods    base

other attached packages:
[1] tidyr_1.3.2      tibble_3.3.1      scales_1.4.0      readr_2.2.0
[5] knitr_1.51       gt_1.3.0          ggthemes_5.2.0    ggplot2_4.0.3
[9] ggthemes_5.2.0    dplyr_1.2.1       asar_2.5.0.9000    afscOSA_0.0.1

loaded via namespace only:
[1] generics_0.1.4      xml2_1.6.0          lattice_0.22-9      stringi_1.8.7
[5] hms_1.1.4           digest_0.6.39       magrittr_2.0.5      evaluate_1.0.5
[9] grid_4.6.1          RColorBrewer_1.1-3 fastmap_1.2.0       plyr_1.8.9
[13] jsonlite_2.0.0      Matrix_1.7-5        tinytex_0.60        purrr_1.2.2
[17] viridisLite_0.4.3   cli_3.6.6           rlang_1.3.0         crayon_1.5.3
[21] litedown_0.9        commonmark_2.0.0    cowplot_1.2.0       bit64_4.8.2
[25] withr_3.0.3         yaml_2.3.12         otel_0.2.0          tools_4.6.1
[29] parallel_4.6.1      reshape2_1.4.5      tzdb_0.5.0          png_0.1-9
[33] vctrs_0.7.3         R6_2.6.1            lifecycle_1.0.5     stringr_1.6.0
[37] fs_2.1.0            bit_4.6.0           vroom_1.7.1         pkgconfig_2.0.3
[41] pillar_1.11.1       gtable_0.3.6        Rcpp_1.1.2          glue_1.8.1
[45] xfun_0.60           tidyselect_1.2.1    farver_2.1.2        htmltools_0.5.9
[49] labeling_0.4.3      rmarkdown_2.31      TMB_1.9.21          compiler_4.6.1
[53] S7_0.2.2            markdown_2.0

```

A-2 Rceattle-to-spmR projection bridge

The projection bridge is a deterministic file adapter rather than a second assessment fit. The [input writer](#) reads the saved `nonparametric_pm` Rceattle object, extracts the population state, biological schedules, fishery schedule, and historical series needed by the Standard Projection Model (SPM), and writes them in the SPM `pm.prj` format. It also writes the projection controls in `spm.dat`. The [projection runner](#) passes those files to `spmR::runSPM(..., engine = "admb")`, which executes the validated SPM binary and returns the simulation-level output used by the report.

The adapter preserves the terminal assessment state and makes each conversion explicit. Numbers at age, natural mortality, and the weight schedules are passed on their Rceattle scales. With abundance in millions of fish and weight in kilograms, their product has units of thousand metric tons. Maturity is divided by its maximum, matching the SPM input convention used for this stock. The spawning-month encodings differ by one because Rceattle records elapsed months and SPM uses a one-based calendar month; both represent survival through 0.25 of the year (Table [19](#)).

Table 19: Mapping from the saved Rceattle fit to the SPM species input file.

SPM field	Rceattle source	Transformation and years
Average recent F	quantities: F_spp	Arithmetic mean for 2020–2024; used by Alternative 3
Natural mortality at age	quantities: M_at_age	Terminal-year schedule (2024)
Maturity at age	data_list: maturity	Divided by its maximum to match the SPM convention
Spawning weight at age	data_list: weight and ssb_wt_index	Terminal-year spawning-weight schedule
Fishery weight at age	data_list: weight, fishery index	Terminal-year fishery-weight schedule
Fishery selectivity at age	quantities: sel_at_age for Fishery	Arithmetic mean at each age for 2020–2024
Terminal numbers at age	quantities: N_at_age	Terminal assessment state (2024)
Recruitment history	quantities: R	Annual age-1 recruitment for 1978–2024
Spawning-biomass history	quantities: ssb	Annual spawning biomass for 1977–2023
Spawning month	data_list: spawn_month = 3	Written as SPM Spawnmo = 4; both equal 0.25 year

Annual Rceattle fishery selectivity is already normalized to a mean of one across ages. The projection schedule is calculated as the arithmetic mean at each age across 2020–2024; averaging these normalized annual curves retains a mean of one. This five-year schedule replaces the former terminal-year-only curve. The writer saves the five contributing curves in `fishery_selectivity_recent_years.csv`, the applied average in `age_schedules.csv`, and the rule and exact years in `lineage.csv`. Thus the projection input can be reconstructed without reading values back from the formatted `pm.prj` file.

The `spm.dat` file starts projections in 2025, runs 14 projection years and 1,000 simulations, and requests all seven Tier 3 alternatives. Catch is fixed at 1,350 thousand t in 2025 and 2026 before the alternatives diverge. A separate call uses Alternative 2 with catch fixed at 1,300 thousand t for 2025–2032. The writer copies the validated SPM executable and `tacpar.dat` control file into each run directory. `manifest.csv` records the role and presence of every required input, while `lineage.csv` records the terminal year, software versions, selectivity rule, and averaging years.

The saved products passed all file and field consistency checks. These checks do not replace review of the harvest assumptions or the biological appropriateness of a five-year selectivity average (Table 20).

The means formerly displayed in the long main-text table remain available as a [downloadable CSV](#). This file retains every alternative and projection year in a compact, machine-readable form. The underlying simulation-level output remains in the saved analysis products but is omitted from the report because of its size.

Table 20: Automated consistency checks for the Rceattle-to-spmR projection bridge.

Check	Result	Evidence
Required projection files	Pass	6 of 6 manifest targets present
Recent selectivity years	Pass	2020, 2021, 2022, 2023, 2024
Applied five-year selectivity mean	Pass	maximum absolute difference = 3.996803e-15
Projection alternatives	Pass	1, 2, 3, 4, 5, 6, 7
Projection start year	Pass	2025

A-3 Draft EBS pollock ESP development framework

The most effective way to integrate the ecosystem work into this working paper is to keep the concise DSEM pilot in Section 7 and develop the full ESP as an appendix. This preserves the assessment narrative while giving the ESP its own standard structure, evidence trail, and update cycle. The scaffold below adapts the national ESP tables and the 2026 Gulf of Alaska arrowtooth flounder ESP ([Alaska Fisheries Science Center Ecosystem and Socioeconomic Profile Coordination Team 2026](#); [Shotwell et al. 2026](#)). It is a September draft: the indicator synthesis and proposed causal structure are present, while the final indicator updates, expanded candidate-DAG comparison, and management summary are November deliverables. The proposed components and their present status show which elements are complete and which remain scheduled for November (Table 21).

Table 21: Proposed EBS pollock ESP appendix structure and status in this working paper.

ESP component	EBS pollock content	Current status
Executive summary	ABC/TAC relevance, predictive and contextual indicators, principal data gaps.	Write after the November indicator analysis.
Introduction and justification	2021 BSAI Plan Team request; focal response is age-1 recruitment and its early-survival mechanisms.	Justification documented; team review pending.
Data	Assessment states, physical indices, plankton and prey indices, age-0 condition, predation/overlap products, and selected socioeconomic series.	Inventory started; several proximal biological series remain short or intermittent.
Indicator synthesis	Life-history table, conceptual diagram, generic DAG, and mechanism table developed from the workshop and working-group material.	Draft synthesis presented here and in Section 7.
Proposed indicators	A small set selected by mechanism, timing, record length, uncertainty, and availability rather than correlation alone.	SST and cold pool illustrate the workflow; age-0 energy and lipid-rich prey are priorities.
Indicator assessment	Current condition, trends, uncertainty, and interpretation for ecosystem and socioeconomic indicators.	Complete after current-year ESP and ESR data collection.
Indicator analysis	No-edge reference, candidate DAGs, common-data AIC, path estimates, residual-variance change, predictive checks, and scientific sign checks.	Two proxy examples completed; expanded candidate-DAG analysis deferred to November.
Data gaps and priorities	Fall age-0 energy, prey lipid/abundance, bloom-prey match timing, juvenile-adult overlap, and consistent observation-error information.	Priorities identified; feasible annual products require team confirmation.
Conclusion and management use	Separate predictive indicators from contextual indicators and state whether each is suitable for recruitment modeling, the risk table, or monitoring only.	Complete after validation and team review.

The draft life-history synthesis condenses the standard ESP table to the stages and information needed for the recruitment question (Table 22). A later stand-alone ESP can expand each stage into the national-template fields for habitat and distribution, phenology, growth, energetics, diet, and predators or competitors.

Table 22: Draft life-history synthesis for EBS pollock recruitment, condensed from the national ESP template.

Stage	Process emphasized	Available evidence	Priority gap
Spawning and eggs	Adult condition, spawn timing and location, egg development, and transport.	Adult condition and physical indices are regularly available.	Cohort-specific egg production and transport exposure.
Larvae	Bloom timing, prey match, transport, temperature-dependent growth, and predation.	Physical forcing is available; biological observations are less consistent.	Annual bloom-prey match and larval survival indices.
Young of year	Prey quantity and lipid quality, growth, distribution, and accumulation of fall energy.	SST is long; prey and condition series are shorter or intermittent.	Consistent fall age-0 abundance, diet lipid, energy density, and total energy.
Overwinter age 0 to age 1	Energy reserves, metabolic demand, winter severity, and predation determine survival.	Physical proxies and modeled recruitment are available.	Direct annual overwinter-survival observations and uncertainty.
Juvenile and pre-recruit	Spatial overlap with adults and other predators affects cannibalism and natural mortality.	Adult abundance is long; overlap and mortality products require timing review.	Cohort-aligned juvenile-adult overlap and predation mortality.
Adult	Spawner abundance, condition, distribution, and cannibalism affect production and juvenile survival.	Assessment abundance, maturity, distribution, and condition products are available.	Separate maternal and density-dependent pathways from shared assessment information.

The generic DAG in Section 7 serves as the initial full conceptual graph. The fitted SST and cold-pool graphs are deliberately simplified descendants of that graph, not a search across every available series. For November, each candidate graph should be traceable to a row in the mechanism table and should use the same analysis data block as the IID no-recruitment-edge reference. The final ESP should then report the causal statement, lag, expected sign, data limitations, path estimate, uncertainty, change in unexplained recruitment variation, and predictive check for every retained recruitment edge.

Fishery performance and socioeconomic indicators can be transferred from the SAFE after the biological indicator set is stable. They should occupy the standard socioeconomic synthesis and monitoring subsections rather than be inserted as recruitment causes unless a separate, defensible causal hypothesis requires them. This division keeps the integrated Rceattle-DSEM analysis focused on the population response while allowing the ESP appendix to inform both biological risk and TAC considerations.

A-4 Model fitting details

The fishery selectivity likelihood has multiple modes, so the reported Rceattle configuration used data-based fishery-selectivity starting values and two fitting stages. The first stage disabled the annual selectivity deviations to establish the selectivity and abundance scale. The second stage restored the deviations and used the first-stage solution as its initial values. This sequence produced a small maximum gradient and fully met the convergence criteria. With the direct ATS/AVO coefficient-increment penalty, the Hessian condition number was approximately 489,000; its weakest direction remained in the shared ATS/AVO annual selectivity deviations (`sel_coff_dev`) rather than fishery/CPUE.

The complete fitting sequence, including the empirical starting values and the two estimation stages, is in the [two-stage fitting script](#) and is run with Rscript "2024 EBS pollock.R". The nine-peel implementation is in the [retrospective script](#) and is run with Rscript "2024 EBS pollock two-stage retrospectives.R".

Table 23: Disposition of the Rceattle development components.

Item	Development component	Review result	Interpretation
1	Prepared Rceattle model-input workbook	Used the delivered workbook directly.	Input files are available and reproducible from stored files.
2	Rceattle fit and 2D AR1 sensitivity	The Rceattle fit completed; 2D AR1 received a FAIL convergence status.	The NonParametricPM configuration is the relevant Rceattle comparison; 2D AR1 is diagnostic only.
3	Convergence and model diagnostics	Structured convergence checks, OSA diagnostics, nine retrospective peels, 50 self-tests, and 50 jitters are saved.	The ATS/AVO coefficient-increment penalty resolves the prior Hessian-conditioning warning; two inferior jitter solutions remain.
4	Next-assessment data update workflow	Update script review is complete; execution awaits complete new-year observations.	Useful as a template only after complete new-year observations are inserted.
5	ADMB-to-Rceattle comparison diagnostics	Setting Rceattle to the ADMB values reproduces ADMB to rounding-scale precision.	This is the strongest evidence that the ADMB comparison is correct.

The retrospective peels repeat this sequence after recalculating the data-based selectivity starting values for each terminal year. Because `sel_coff_dev` contains annual random-walk increments at age, each peel fixes the terminal-year fishery and mirrored catch-per-unit-effort increments at zero. This holds their terminal selectivity-at-age curves at the preceding-year curves. The completed development components distinguish the accepted comparison, diagnostic work, and unresolved research sensitivities (Table 23).

A-5 Jitter and natural-mortality profile

The [extended-diagnostics runner](#) generates the 50 jitter fits with `Rscript R/run_rceattle_5_8_1_extended_diagnostics.R`.

The 50 phased jitter runs test sensitivity to dispersed starting values. Forty-eight returned to the best objective within numerical precision. Two received `WARN` status and reached an inferior mode with an objective increase of approximately 609.5. This reproduces the pre-existing multimodality and shows why convergence of one run is not sufficient evidence that the global mode was found. Forty-eight runs returned to the best mode and two reached the inferior mode (Table 24).

The age-3-and-older natural-mortality profile was evaluated from 0.15 to 0.50. The saved profile fits retain Rceattle's full negative-log-likelihood (NLL) component matrix. Each component in Figure 15 is shown as delta NLL relative to its value at the grid minimum of 0.20. The total-objective curve equals the sum of the data and penalty components. The shallow total profile indicates that these data provide limited information

Table 24: Rceattle 5.8.1 phased-jitter results for the fitted configuration. Delta objective is measured from the best of 50 runs.

Status	Runs	Minimum delta objective	Maximum delta objective
Criteria fully met	48	0.000	0.000
WARN	2	609.548	609.548

for estimating natural mortality.

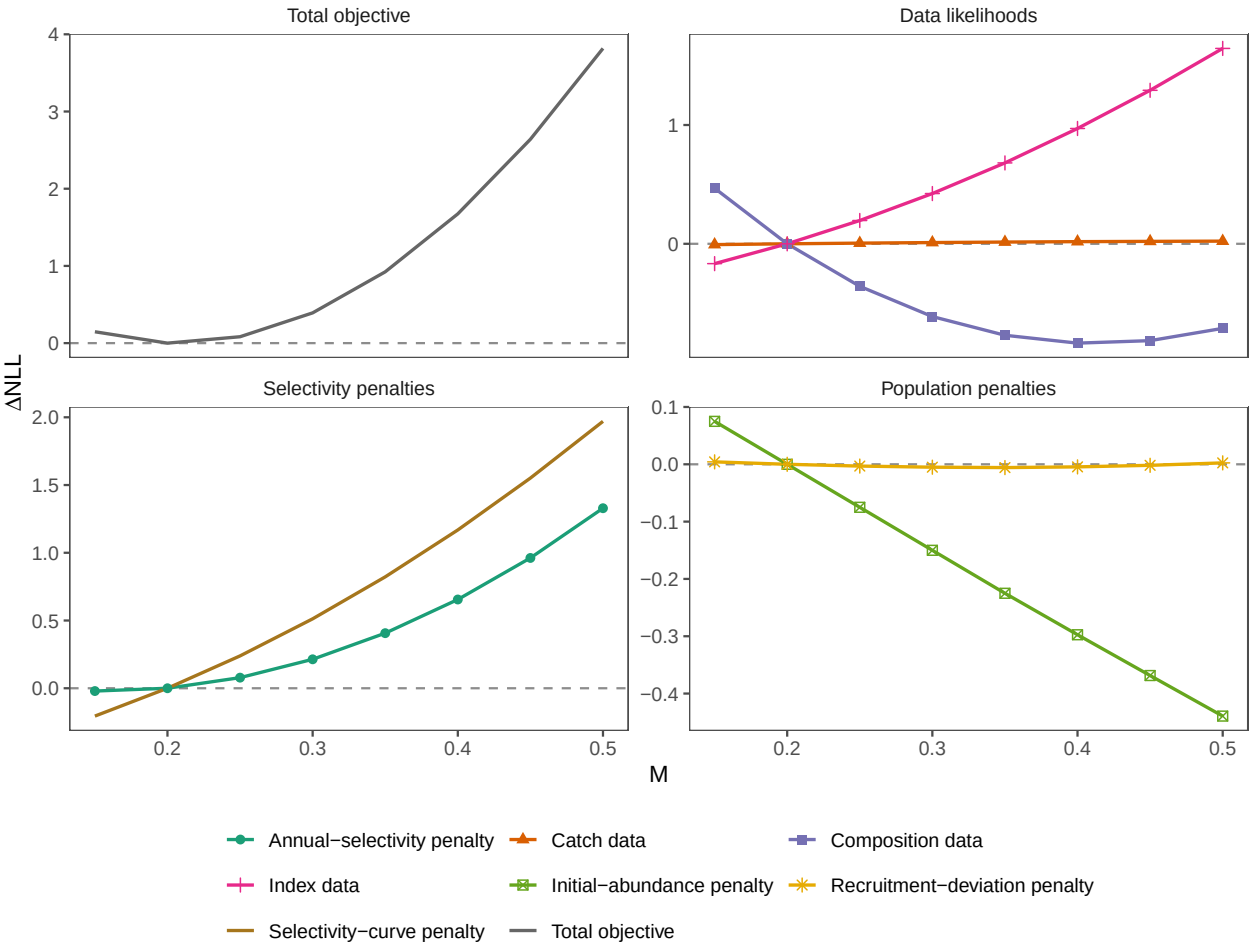


Figure 15: Negative-log-likelihood components across the age-3-and-older natural-mortality profile. Each curve shows delta NLL relative to that component’s value at $M = 0.20$. Positive values favor $M = 0.20$ over the plotted value; negative values favor the plotted value. Components in the three lower panels sum to the total objective change in the upper-left panel.

A-6 2D age-by-year AR1 selectivity sensitivity

The penalized-effects initialization and subsequent Laplace fit are in the [two-stage fitting script](#) and are run with `Rscript "2024 EBS pollock.R"`. A separate [fixed-scale follow-up](#) can be run with `Rscript "2024 EBS pollock 2D AR1 fixed sd test.R"`; it is a development check and is not the sensitivity summarized below.

The 2D AR1 sensitivity applied an age-by-year field to the shared fishery and catch-per-unit-effort selectivity block over 1964–2024. The age-year deviations were first fit as penalized effects to initialize the field and then integrated as random effects with a Laplace approximation. The initialized Laplace fit failed the gradient, Hessian, and bound checks. The estimated correlation and scale parameters are retained only as diagnostics (Table 25).

Table 25: Diagnostic 2D AR1 fishery-selectivity correlation and scale estimates.

Quantity	Rceattle parameter	Penalized effects	Random-effects fit
Year AR1 correlation	Sel_curve_pen1	0.584	0.947
Age AR1 correlation	Sel_curve_pen2	0.120	0.098
Selectivity deviation scale	sel_dev_log_sd	0.500	0.379

Correlations are on the -1 to 1 scale; the deviation scale is a positive standard deviation. The fit failed convergence checks.

The failed random-effects fit had a maximum marginal gradient of 19.7 on `rec_pars`, a non-invertible Hessian, 61 `log_F` parameters at their configured bounds, and **FAIL** convergence status. Its selectivity surfaces differ visibly from both the ADMB reference and accepted Rceattle configuration, but the failed fit precludes biological interpretation (Figure 16).

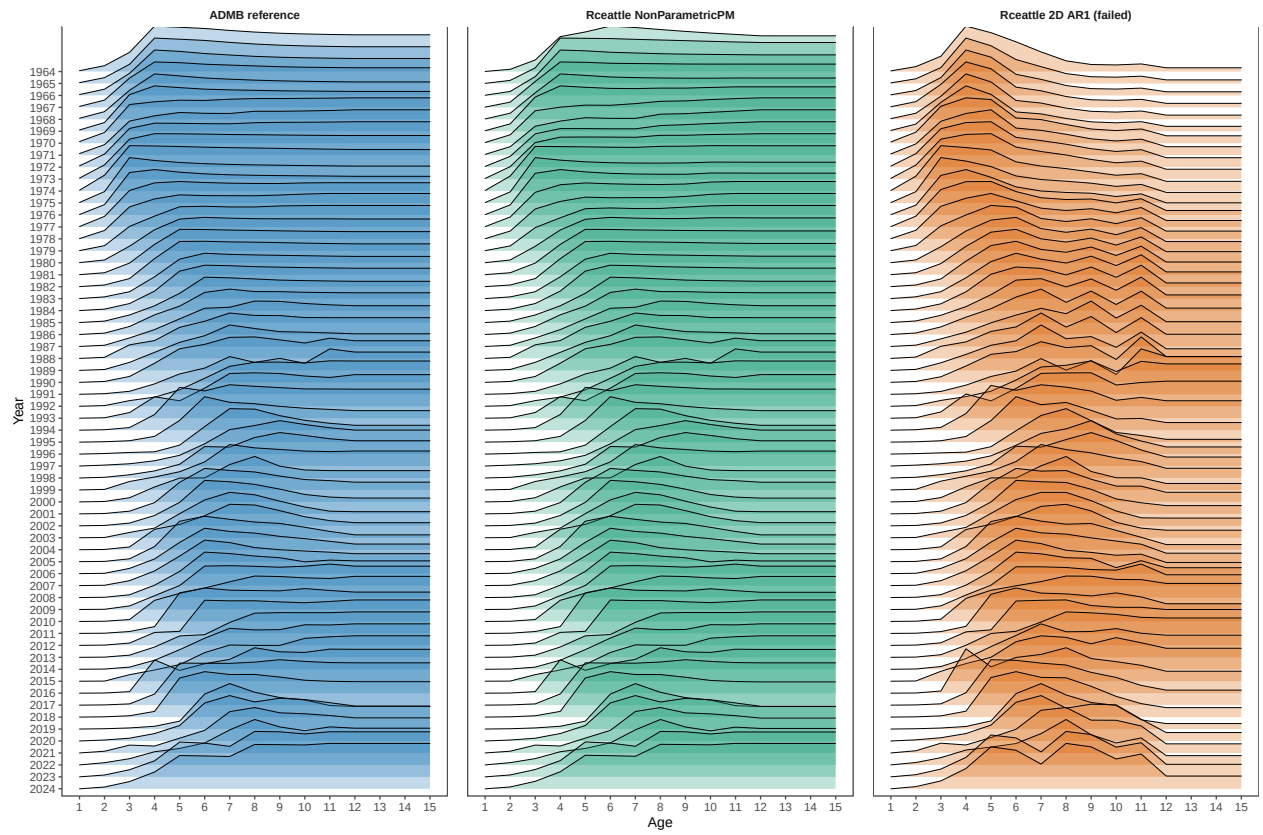


Figure 16: Development comparison of relative fishery selectivity for ADMB, the Rceattle configuration, and the failed 2D age-by-year AR1 sensitivity.

A-7 Observation continuity and case-deletion influence

This diagnostic separates whether an observation was available and used from how much the accepted fit changes when a fitted sampling event is removed. The observation-continuity panel (Figure 17) uses an explicit assessment-input schedule: annual for the fishery, BTS, and AVO rows and even years for ATS rows. Years after 2024 are shown only to distinguish the current fitted span; the schedule does not imply a commitment about future field operations.

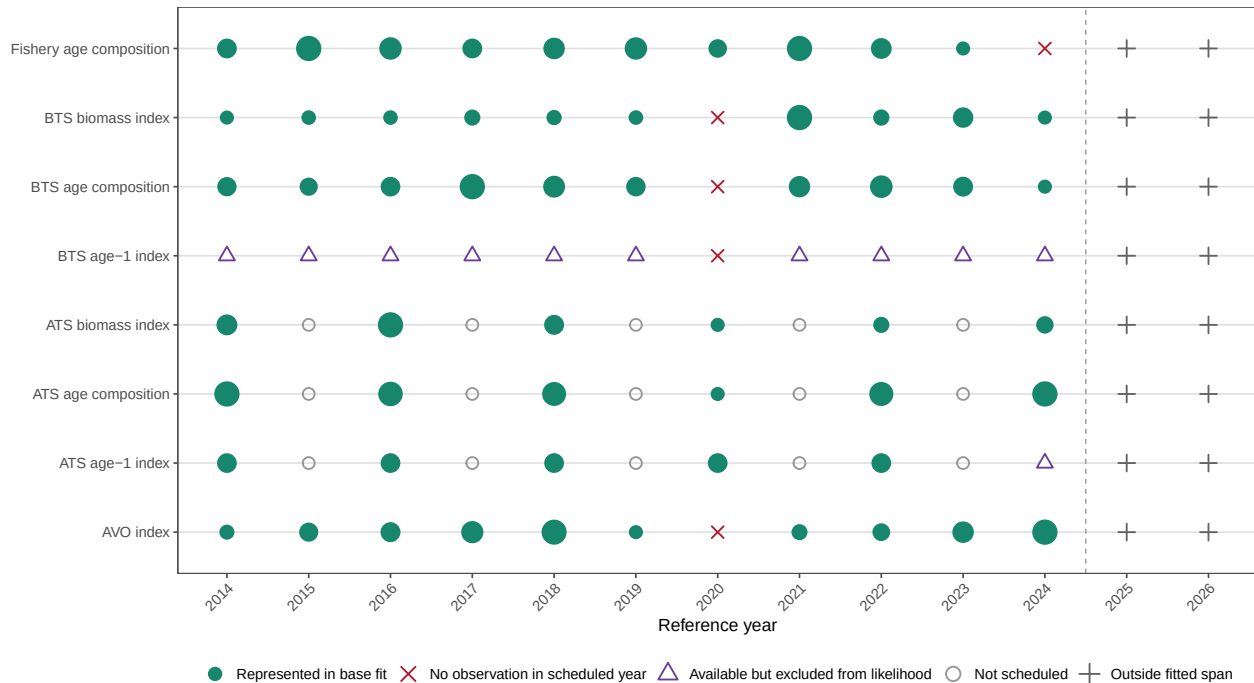


Figure 17: Observation continuity in the accepted EBS pollock Rceattle configuration, 2014–2026. Filled green circles are observations represented in the base fit. Red crosses identify scheduled years with no observation, purple triangles identify observations supplied but excluded from the likelihood, open circles are years outside a component’s sampling schedule, and gray plus signs are outside the fitted span. Bubble area is scaled within, not among, components: nominal sample size for composition rows, inverse diagonal covariance for the BTS biomass index, and inverse squared observation standard deviation for the other indexes.

The standalone BTS age-1 series is present in the input workbook but its fleet control is `Off`; those observations are therefore not represented in the likelihood. The ATS age-1 index is active through 2022, whereas its 2024 row is negative-year coded and excluded. The fishery age-composition series ends in 2023, leaving a scheduled 2024 gap. The 2020 BTS and AVO gaps are also visible, while odd ATS years are correctly classified as not scheduled rather than missing.

Each composition deletion removed the complete fleet-year row, retaining its age bins as one multinomial sampling event. Each index deletion removed one fleet-year row. For the BTS biomass index this is an explicit case deletion from the joint covariance likelihood, not a claim that Cholesky-whitened OSA terms are independent annual likelihoods. The 12 controlled refits retained the accepted parameter map, model controls, robustifying constant ($c = 0.001$), and full-data solution as their starting point. The [diagnostic runner](#) writes a [continuity inventory](#), [refit manifest](#), [summary](#), assessment-year trajectories, and objective-component totals. The terminal changes are displayed in Figure 18 and summarized in Table 26.

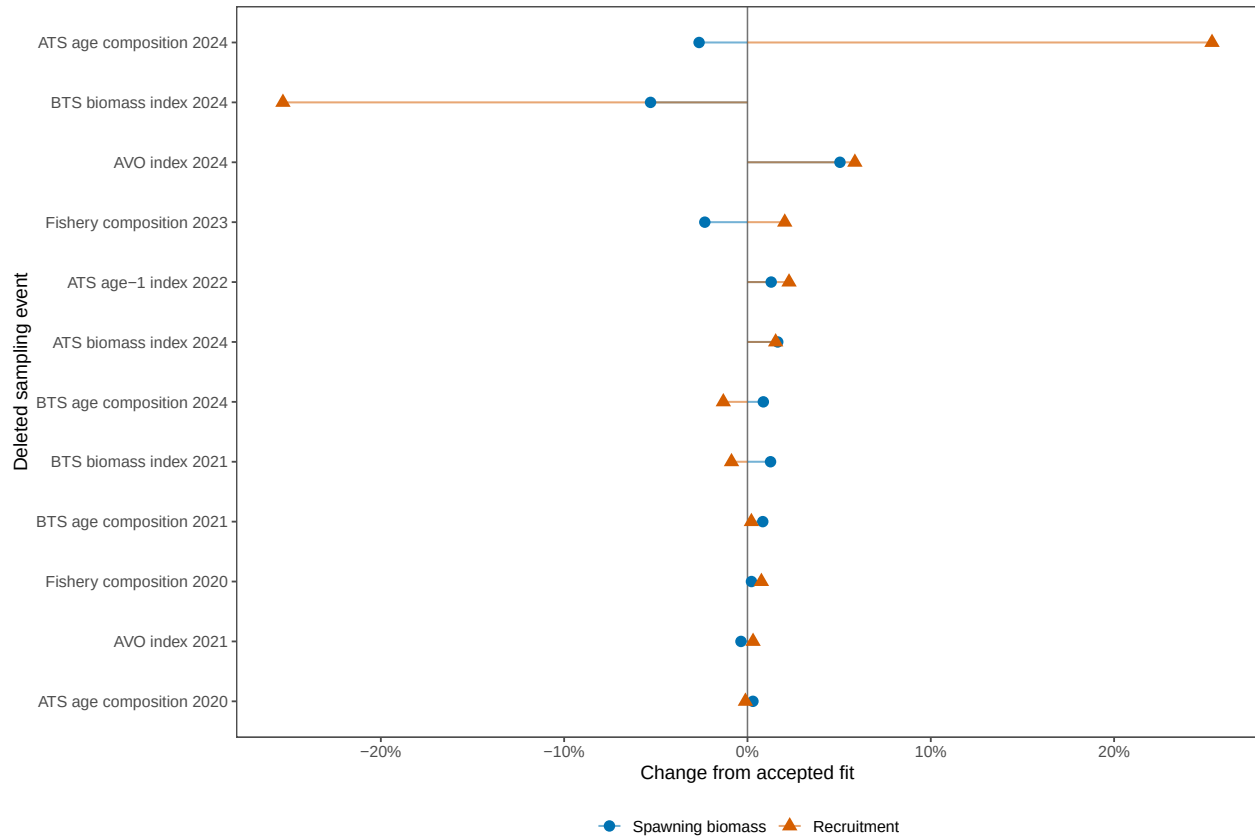


Figure 18: Case-deletion changes in 2024 spawning biomass (SSB) and age-1 recruitment relative to the accepted full-data fit. Each point is a complete event-level refit; positive values indicate a larger estimate after deleting the event.

Across Figure 18 and Table 26, all 12 refits had status OK, positive-definite Hessians, and maximum absolute gradients below $8.2\text{e-}05$. The largest terminal SSB response was -5.29% after deleting the 2024 BTS biomass index. The largest terminal recruitment responses were approximately 25.3% in opposite directions: lower without the 2024 BTS biomass index and higher without the 2024 ATS age composition. The largest recruitment response in any assessment year was 53.9%, for the 2023 recruitment estimate when the 2024 ATS age composition was removed. These are sensitivity measures, not evidence that an observation is erroneous or should be down-weighted.

Table 26: Case-deletion results. Maximum changes are evaluated over assessment years through 2024.

Deleted event	2024 SSB change (percent)	2024 recruitment change (percent)	Maximum recruitment change (percent)
Fishery composition 2020	0.22	0.76	14.50
Fishery composition 2023	-2.33	2.03	11.81
BTS biomass index 2021	1.26	-0.87	1.39
BTS biomass index 2024	-5.29	-25.34	25.34
BTS age composition 2021	0.83	0.21	7.48
BTS age composition 2024	0.87	-1.32	15.69
ATS biomass index 2024	1.65	1.54	2.46
ATS age composition 2020	0.30	-0.11	0.33
ATS age composition 2024	-2.64	25.34	53.92
ATS age-1 index 2022	1.29	2.27	8.64
AVO index 2021	-0.35	0.31	0.56
AVO index 2024	5.04	5.86	9.79

The two strongest cases in Figure 18 and Table 26 were also refit from the saved full-data stage-A estimates. Their alternate and primary objectives differed by at most $1.4\text{e-}08$, and their terminal changes agreed within 0.001 percentage points. This targeted check supports the numerical stability of the two largest reported effects.

The objective differences saved with these diagnostics are descriptive only: the fits contain different observations and the accepted stabilized composition objective is a pseudo-likelihood, so they are not likelihood-ratio tests. Formal PSIS-LOO remains a separate research task. It requires a proper grouped data density, a validated posterior sample, and conditional treatment of annual observations in the BTS covariance likelihood (Vehtari et al. 2017, 2024). The existing [SparseNUTS analysis](#) remains exploratory and is not used for assessment inference.

A-8 Acronyms

This appendix follows the glossary convention in the [Alaska Stock Assessment Report \(asar\)](#) package (Schiano et al. 2026). The project glossary uses ASAR’s `\newacronym` format and adds terms specific to this Rceattle comparison. The quoted label “PM” is a model name. The glossary defines the abbreviations used throughout the report (Table 27).

Table 27: Acronyms used in this working paper.

Acronym	Definition
ADMB	AD Model Builder
AIC	Akaike information criterion
AR1	first-order autoregressive
ATS	acoustic-trawl survey
AVO	acoustic vessels of opportunity
BTS	bottom trawl survey
CIE	Center for Independent Experts
CPUE	catch per unit effort
CSV	comma-separated values
DAG	directed acyclic graph
DSEM	dynamic structural equation modeling
ESP	ecosystem and socioeconomic profile
HTML	HyperText Markup Language
IID	independent and identically distributed
LOO	leave-one-out cross-validation
MCMC	Markov chain Monte Carlo
MVN	multivariate normal
OSA	one-step-ahead
PSIS	Pareto-smoothed importance sampling
RDS	R serialized data
RTMB	R Template Model Builder
SAFE	Stock Assessment and Fishery Evaluation
SDNR	standard deviation of normalized residuals
SEM	structural equation model
SPM	Standard Projection Model
SST	sea surface temperature